8

AI in Creative Industries and Media

Image · sound · video · 3D · recognition · wrap-up



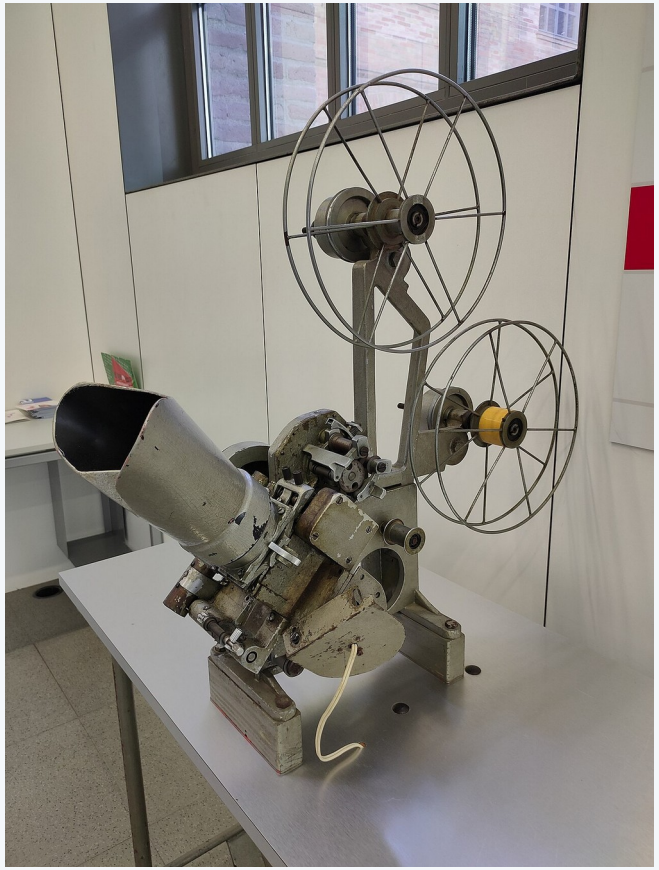
Over sixty years the tool got cheaper seven times, and each time the intermediary operation disappeared

20th century

1961 — ink tracing; staff halved → 1990 — Scitex; $300/hour vs. $895 forever → 1989–2003 — splicing film; a $50–80K workstation → 1997 — a retake for one note; pitch correction

21st century

2000s — shoot-for-catalog on spec; from hundreds of dollars to $0.25 → 2002 · 2009 — hand-animated faces; capture became measurement → 2019 — a location shoot for a backdrop; 1,326 panels, a 270° arc



Left to right — time; the second row continues the first. The intermediary operation disappeared; the profession stayed

The turning point — the 2000s. Unit price had already fallen three orders of magnitude here — the market survived, splitting into "cheap and plentiful" and "expensive and exactly this."

Not one of the seven lines removed the profession that disappeared.

Disney, "101 Dalmatians," 1961 · Adobe Photoshop 1.0, 02/19/1990 · Avid Media Composer, 1989–2003 · Auto-Tune, 09/19/1997 · iStock 2000 and Shutterstock 2003 · "The Two Towers" 2002 and "Avatar" 2009 · ILM StageCraft, 2019

In 2022, what got cheaper wasn't execution — it was the failed attempt.

THE CLASSIC MODE OF THE SAME SPOT

a crew of 50+ people, a year's schedule

named by the company's own leadership

WITH A MODEL, 2025

70,000 clips, 5 specialists, 30 days

the campaign itself still ran on roughly the same 100 people

WHAT CHANGED

the number of trials, not the number of people

this is exactly the measured fact of the row

Without a model	idea	a trial costs as much as the result: $50–150 and 1–3 days	two or three options get looked at
With a model	idea	**a trial costs $0.034–0.05 and seconds**	hundreds get looked at — there is no one to choose among them

Manolo Arroyo, Coca-Cola, The Wall Street Journal · implicator.ai, "Coca-Cola debuts AI holiday ads," 2025 · Toys "R" Us Studios × Native Foreign, June 2024

One mechanism, three scales

AN IMAGE

$0.034–0.05

classic mode: $50–150 per illustration, 1–3 days

A MINUTE OF SPEECH

$0.03–0.072

classic mode: a voice talent, $50–450 per minute

AN EPISODE OF DRAMA

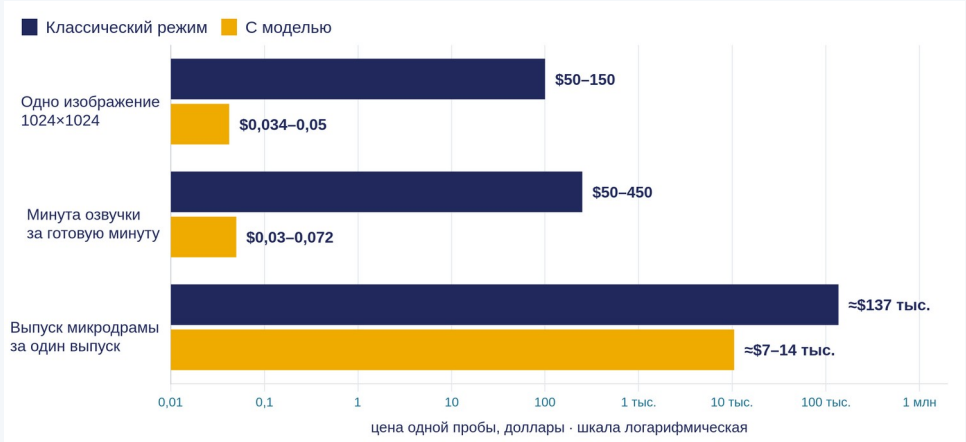
≈$7–14K

classic mode: ordinary production, ≈$137K

THE GAP ACROSS ALL THREE

×10–20 at the expensive end, ×1000–4500 at the cheap end

the cheaper the unit, the bigger the gap



There is one unit on this frame: the price of one trial.

Three orders of magnitude of budget. One mechanism: paying for a trial stopped.

ai.google.dev/gemini-api/docs/pricing (read 10/27/2026) · elevenlabs.io/pricing/api (10/27/2026) · Caixin Global, "How AI took over China's micro-drama industry," 06/01/2026

Execution stopped being the scarce resource. What's scarce now — and in what units is it paid for?

Execution stopped being the scarce resource.

What's scarce now — and in what units is it paid for?

The map of the evening: six task types, one column added per section

The rows are set up front. A column gets added at the end of each section.

Тип задачи	*Image*	*Sound*	*Video*	*3D*	*Recognition*
Generate a first approximation *get candidate looks while the decision is still open*					
Bring to production quality *select, finish, and ship what's good*					
Restore what already exists *pull out something already recorded but hard to read*					
Hold exact measurements *dimensions, tolerances, checkable conditions*					
Fit the real-time budget *meet the frame's deadline, not the task's deadline*					
Answer 'who is accountable' *provenance, rights, the signature behind the result*					

PAYMENT ADDRESS накапливается по разделам: selection · rights · discernment

The path from request to result: one figure, five sections ahead

SAFETY, FORGERY, ABUSE

Lecture 14

deepfakes in personnel security, voice fraud, document forgery, law enforcement

AGENTS, TOOLS, PROGRAMMATIC ACCESS

Lecture 3

call chains, adapter fine-tuning, a human validator — taken as already known

THE LABOR MARKET AND A COUNTRY-BY-COUNTRY JURISDICTION REVIEW

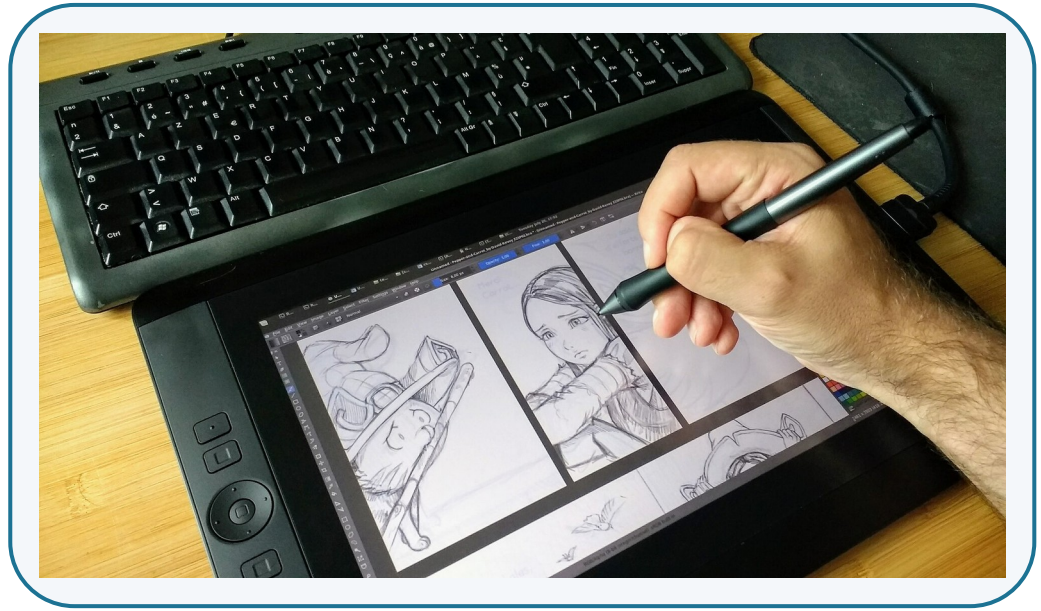
in the chapter, as reading material

collected and checked, but does not go onto the slides: it would crowd out the capabilities

1. request → 2. how it gets inside → 3. where the result starts

4. the step, repeated N times → 5. how the visible thing is produced → 6. result

Section 1. Still Image



What kind of work this is: the stage, the classics, and two cost drops before any model

THE STAGE'S QUESTION

"what does it look like"

not "what is it" (a drawing) and not "how much of it" (a table)

CLASSICS: FOUR TIERS

freelancer $50–150 · catalog shoot — hundreds of $ · microstock $0.25–1 · in-house staff — no price

a catalog shoot isn't replaced by generation; an in-house designer has almost no published numbers

TWO COST DROPS BEFORE THE MODEL

xerography 1961 — headcount ×0.5 · Photoshop 1990 — $300/hr → $895 forever

the intermediary operation disappeared, not the profession; output volume grew faster than the price fell

THE SECTION'S THREE PAYMENT ADDRESSES

selection · rights · discernment

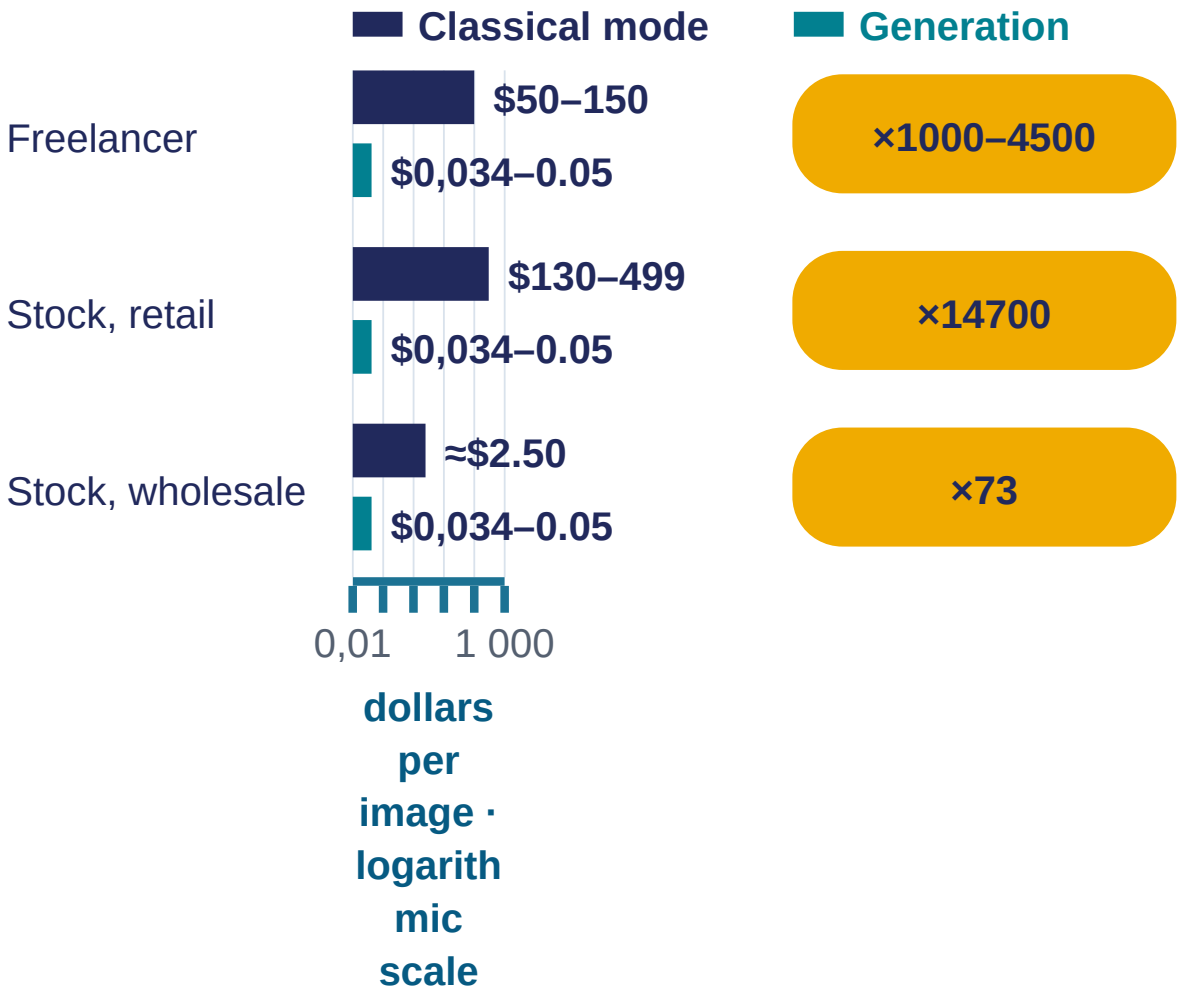
we pay in detail for the first two; discernment comes as a mention, in the section on voice

The stage has one decision — approve the look before the big spending starts. Before 2022, the classics ran through four tiers: in every case what got cheaper was executing the decision, not the choice itself.

The section pays for selection, rights, discernment.

Profi.ru, a sample of 3,848 Moscow illustrator profiles · analysis of a thousand Upwork profiles · stock agency price lists, iStock 2000, Shutterstock 2003 · Disney, "101 Dalmatians," 1961 · Adobe Photoshop 1.0, February 19, 1990

Case: The Collapse of the Per-Image Unit Price





ai.google.dev/gemini-api/docs/pricing (read 27.10.2026) · profi.ru, 3,848 Moscow illustrator profiles · stock agency price lists · Toys «R» Us Studios × Native Foreign, June 2024

Case: T—Zh Trained Its Own Model on a Home GPU

FINE-TUNING

1.5 hrs on an RTX 3080 (16 GB, laptop)

40 images on top of the base model

OUTPUT

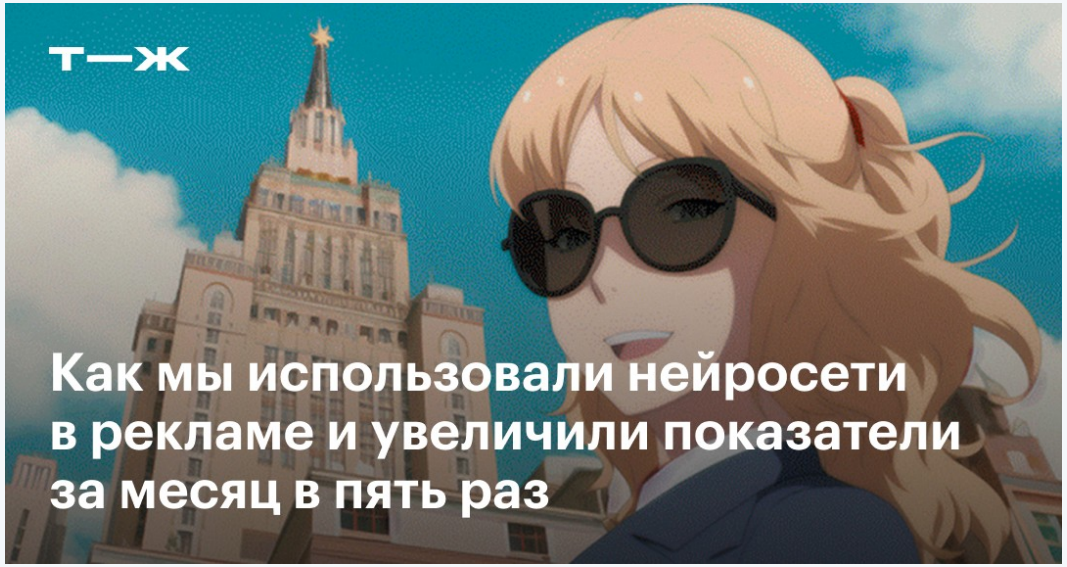
46 posters in February

before that — 23 visualizations

SELECTION

30–100 generations, ≈0.5 hr per illustration

12 different models tried



HIT RATE

2/3 of banners — wrong click-through rate

the best performed 2–3× better than average

CLASSICAL BASELINE

not published

a designer is named; no price or turnaround given

Payment address: SELECTION.

Dropped before the breakdown. An industry overview credits the editorial team with abandoning stock photos for Kandinsky; in fact it's Midjourney and Stable Diffusion, and the subject is banners.

"Tinkoff Journal," "How We Used Neural Networks in Advertising," March 20, 2023

Case: Airbus's Bionic Partition — the Win Isn't from a Trained Model

BEFORE: CALCULATED PARTITION

65 kg

the only alternative: the criterion is set by physics

AFTER: GEOMETRY FROM A BONE-GROWTH MODEL

35 kg

−45% mass; Scalmalloy alloy, the largest cabin part at the time

THE PRICE OF THE ERROR — IN THE INDUSTRY'S OWN UNIT

up to 465,000 tonnes of CO_2 a year

Airbus's estimate for the whole A320 order book: every extra kg in the cabin is fuel for the aircraft's whole service life

THE SELECTION CONDITION: REMOVE THE MODEL MENTALLY

nothing disappears — the search, the calculation, the printing all remain

there was no model in the loop; in the T—Zh case, without the model there are no 46 posters left — there the condition passes

The criterion is set by physics: hold the load at minimum mass. There's exactly one alternative, and it's counted in the same unit.

Autodesk News, «Autodesk and Airbus: generative design in the aerospace factory», 2016

Four Observations, Each with a Number

30-DAY DOWNLOADS

461 vs 622,299

Kandinsky 6.0 Pro vs FLUX.1-dev; SDXL — 4,282,392, Wan 2.2 — 3,901

SPEECH AND TEXT FROM THE SAME TEAM

FRIDA 597,351 · GigaAM v3 426,500

three orders of magnitude more than the same country's image models get

MARKUP ON THE EXCHANGE RATE

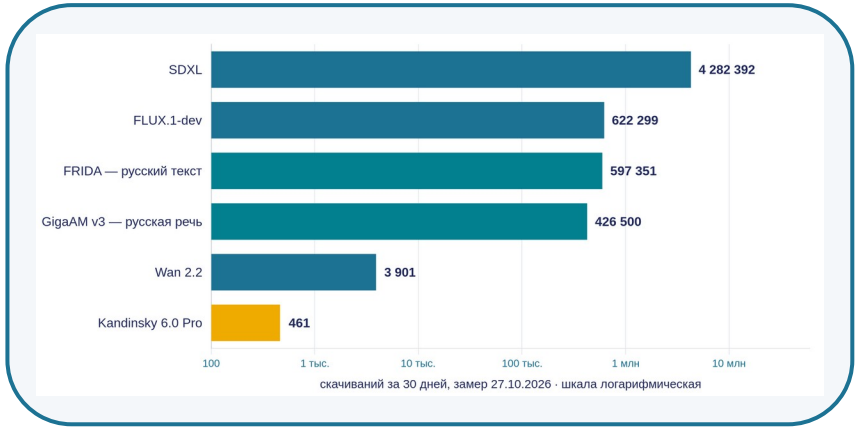
196.42 ₽/$ against 80.49 at the Central Bank

the 2.44× gap matched to the second decimal digit across nine models

NUMBERS ON BEING "LEGALLY CLEAN"

zero

no market share, no attempt counts, no savings, no outcomes



Two Russian services grant opposite rights to one result. One hands the rights to the user, the other keeps them for itself. "Allowed" or "not allowed" is decided by the user agreement.

Hugging Face API, 30-day downloads, queried 27.10.2026 · Kandinsky and Shedevrum user agreements · foundation-models.api.cloud.ru and Cloud.ru's pricing document, revision 23.10.2026 · Adobe's indemnification terms

There Are Two Families of Generation, and Each Generator Is Built from Parts Trained Separately

WHY THERE ARE EXACTLY TWO FAMILIES

piece by piece — or the whole thing, many times over

not discreteness versus continuity: image autoregression (GPT-Image, Nano Banana) has discrete tokens too

input · someone else's model

Text encoder

a frozen model reads the description — SD3 has three, FLUX.1 has two

scaling up the encoder gives more than scaling up the network

→

the work

Denoising network

latent 128×128×16, 1–50 passes

the description is unchanged on every step; there's no history

→

output · bottleneck

Autoencoder

the decoder unfolds the latent into 1024×1024

25.12 dB on 4 channels against 28.62 on 16

Two families: autoregression builds up piece by piece, denoising edits the whole object at once, but many times over.

From the description to the file, three independently trained models are at work — the diffusion network is only the middle one of them.

Ho, Jain, Abbeel, «Denoising Diffusion Probabilistic Models», 2020 · Sohl-Dickstein et al., 2015 · Rombach et al., 2021 · the SD3 technical report, 2024 · Imagen, 2022

The path from request to result: denoising the latent

REPEATS AND THE REQUEST'S ROLE

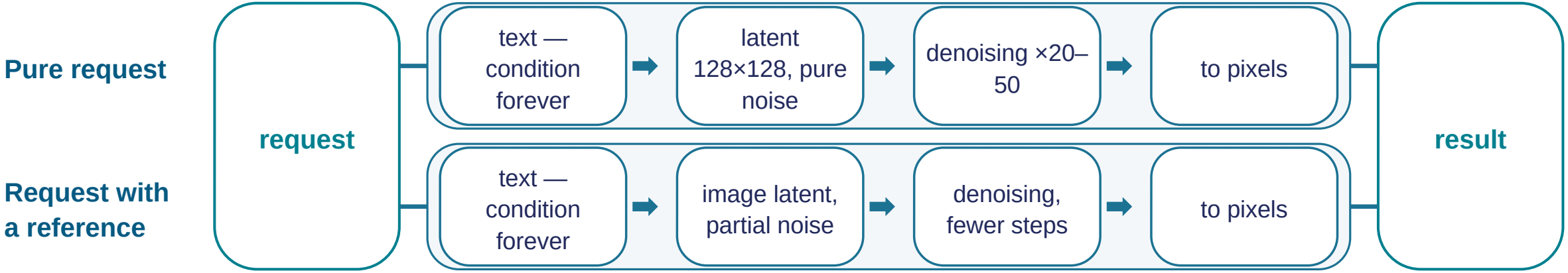
20–50 passes

set by the sampler, not the network size; the text is the condition on every one of them, not just the first; it was 1000 in 2020

WHAT CHANGES, THE BOTTLENECK

the latent — every time; the condition — never

the bottleneck is the number of passes itself: distillation down to 4 delivers the same work 25 times faster



The request goes in once and stays as a condition on each of the 20–50 passes — it's not the request that changes, it's the latent.

Ho, Jain, Abbeel, «Denoising Diffusion Probabilistic Models», 2020 · Rombach et al., 2021 · Ho and Salimans, «Classifier-Free Diffusion Guidance», 2022 · Karras et al., 2022 · Sadat et al., 2024

A 1024×1024 Picture Is 4096 Tokens

WHAT THE AUTOENCODER LOST, MEASURED

25.12 dB on 4 channels against 28.62 dB on 16

perceptual divergence 0.85 against 0.45; this is the ceiling the diffusion network physically can't step past

A DIRECT COMPARISON OF UNITS

there, a token from a vocabulary of ~200k, fixed before training — here, a continuous vector, no vocabulary at all

hence: there's nothing to mask, nothing to forbid

WHAT THIS DELIVERED IN PRACTICE

65,536 numbers instead of 786,432 on every step

one and a half orders of magnitude of compute, on each of dozens of steps — hence 512×512 on a 6–8 GB card in August 2022

pixels: 1024 · 1024 · 3 = 3,145,728 numbers → latent, 16 channels: 128 · 128 · 16 = 262,144 numbers → transformer input, 2×2 patch: 64 · 64 = 4,096 tokens

Three compression stages: area is honestly proportional to the number of values

Quadratic cost hasn't gone anywhere. 4,096 tokens — ≈16.8 million pairs per head per layer, 16,384 tokens — ≈268 million: doubling the picture's side increases attention's cost sixteen times.

The main lever is the tokenizer; model size is secondary here.

Rombach et al., 2021 — the paper Stable Diffusion grew out of · the SD3 technical report, 2024 (the reconstruction-quality table)

The Description Is a Constant Condition on Every Step. And CFG Is Not Temperature

U-Net, SD 1.x–SDXL	Q ← image latent	K, V ← description tokens	map 4,096 × 77 words	it can be drawn out
MMDiT, SD3 and FLUX	text and image together	attention in both directions	map 4,608 × 4,608	the text looks at the image

Temperature

- The model's opinion: doesn't change it — a divisor of the logits
- Direction: higher → more randomness
- Cost: zero — one division
- Randomness: in the sampler, on every step

CFG

- The model's opinion: changes it, pushes past what was predicted
- Direction: higher → LESS diversity
- Cost: two network passes per step
- Randomness: has nothing to do with it — it's in the starting noise

The same Q/K/V mechanism — the query and the keys come from different places.

CFG is not temperature.

U-Net cross-attention · MMDiT joint attention, SD3, 2024 · Imagen, 2022 · Ho and Salimans, «Classifier-Free Diffusion Guidance», 2022 · Sadat et al., 2024

U-Net → DiT → MMDiT → Rectified Flow, and Adapters on Top of Any of Them

THE SLIDE'S MEASUREMENT

XL/8: 676M, 7.39 GFLOPs, Fréchet 106.41 · XL/2: 675M, 118.64 GFLOPs, 19.47

the lever is in how the input was cut up, not in the number of weights

TWO KINDS OF ADD-ONS

LoRA/inversion/DreamBooth — train weights · ControlNet/IP-Adapter — leave them untouched

the same ladder as for a language model; the add-ons stack — style + pose + face on one base

11 CONDITIONS FOR REPRODUCIBILITY

11 items — from the weights to the GPU model

it's different for a language model: 1000 requests to vLLM at T=0 gave 80 different answers (2025)

U-Net, 2015–2023: convolutions + transformer inserts ➡ DiT, Dec. 2022: the latent is cut into patches → a vector ➡ MMDiT, March 2024: text and image — one sequence ➡ rectified flow, 2022→2026: noise and data — a straight line

Four generations of the load-bearing architecture, left to right — time

16 times more tokens — the metric is 5 times better.

Adapters sit on top of the architecture: the base is frozen, the add-ons stack literally.

Ronneberger et al., 2015 (U-Net) · Peebles, Xie, 2022 (DiT) · the SD3 technical report, 2024 (MMDiT) · Liu, Gong, Liu, and Lipman et al., 2022 (rectified flow) · LoRA · textual inversion · DreamBooth · ControlNet · IP-Adapter

Language Model versus Media Model: Architecture, Control — and Cheaper to Train, More Expensive to Run

THE IMAGE MODEL STOPPED GROWING

SD v1 (2022) = SD3 (2024)

stated in their own paper

TRANSPARENCY WORSE THAN LANGUAGE MODELS

2 of 5: SD v1, SD3

SDXL, FLUX.1, DALL·E 3 — not disclosed

	Language Model	Media Model (denoising / flow)
Data unit	token, vocabulary ~200k	patch vector, continuous; **no vocabulary**
Cache	works on an unchanging context	**impossible**: 4096 tokens — every step
What controls the output	temperature, masking — **a guaranteed form**	CFG, adapters — **no guarantees**
Provenance	with retrieval — a citation **exists**	for a generated frame — **none, and there can't be**

The table has 11 rows; seven are in the speaker notes.

Cheaper to train: 150,000 GPU-hours against ≈57 million. **More expensive to run:** $120 per million image tokens against $12 for text — the same model.

OpenAI, Addendum to GPT-4o System Card, 25.03.2025 · Gemini 2.5 Flash Image, 2025 · the SD3 technical report, 2024 · the Stable Diffusion v1 model card · independent compute trackers for frontier models

What the Method Can't Do — and One Property, Read from Two Sides

FAILURE 1 — TEXT

3.5 dB — the line between pseudo-letters and legible text

25.12 dB on 4 channels against 28.62 on 16; without the encoder, long text and complex descriptions degrade

FAILURE 2 — COUNTING

GPT-4o 0.85 against ≈0.99 presence; the best 2026 model — 23.9 of 100

holds even with a vocabulary; the word "three" is an attention weight, not a counter

FAILURE 3 — CONSISTENCY

doesn't exist for diffusion; better for autoregression, not solved

diffusion: no history. Autoregression: growing context, drift over long runs

ONE PROPERTY — TWO SIDES

gave: ×48 compression → a 6–8 GB card, $0,034–0.05 per attempt

took: a letter blurs BEFORE generation, a background face is a handful of latent cells

Letters	48x compression	➡ blurred before the generator	➡ pseudo-letters
Counting	no unit for an object	➡ "three" — an attention weight	➡ nothing to count with

Three failures. Text is the price of the compression that delivered the cheap attempt; counting and consistency hold even with a vocabulary. **The win and the failure come from the same place.**

the SD3 technical report, 2024 (the reconstruction-quality table) · GenEval, GenEval-2 (2026, arXiv:2512.16853), GenOlympiad (2026, arXiv:2610.26850), T2I-CompBench++

There Are No Exact Dimensions: a Generator Doesn't Replace CAD. And "RAG for Pictures" Is a Category Error

THE FORMULATION WORTH CARRYING AWAY

a generator answers "what does it look like." CAD answers "what is it"

substituting the first for the second is a forgery, not a saving

THE SAME CHECK — FOR A TEXT TASK

feeding a source into context works for a language model and for not one single image-generation architecture

for denoising there's no context at all; for autoregression there is context, but no token gets checked against a fact

What you can't do

- Measure it — there's no scale in the image
- Verify it — looks like it passed a calculation, and it didn't
- Manufacture or maintain it: there's no geometry, no build tree

Where a generator is useful

- Look search — storyboarding, shape variants
- Visualizing what's already calculated — rendering, lighting
- The boundary: when a picture becomes a data source

"It looks like an M12 thread" on a picture is not an M12 thread: it's raster points that resemble a photo of a thread.

The technique "a source in context, a citation on the output" doesn't work for pictures — the reasons differ by architecture.

OpenAI, Addendum to GPT-4o System Card, 25.03.2025 · Gemini 2.5 Flash Image, 2025

What Happened to the Image Seller: the Same Maneuver, Twenty Years Apart

2006 — buy the price-breaker. It worked

Event

the price falls — classics $100s–$1000s against microstock $0.25–1

Reaction

the library's owner buys the microstock — its own price-breaker

the upper segment survived — microstock was a NEW CHANNEL for selling the same thing

2026 — buy the competitor. Didn't work

Event

the price falls again — this time because of generation

Reaction

the same player tries to merge with a competitor

generation is the ABSENCE OF A THING: there's no owner, nobody to buy

The channel gets cheaper — whoever survives buys it. The unit loses its value — there's nothing to buy

Subscribers: 321,000 → 240,000 in a year. Debt ≈$2.1 billion, bankruptcy — October 2026.

When a channel gets cheaper, whoever buys it survives. When the unit itself loses its value, there's nothing to buy: it has no owner.

Getty Images financial reports and corporate statements, 2006 and 2026 · stock agency price lists

What It Was Trained On: 1.84 Bytes per Training Image

THE ANSWER

4.27 GB ÷ 2.32 billion images = 1.84 bytes

less than a Cyrillic letter; compression across the corpus ×22,508 — a picture in the corpus weighs 41,379 bytes on average

EVEN BY THE MOST GENEROUS COUNT

26.7 bytes per picture

against 160 million cross-checked examples — fifteen hundred times smaller than a 512×512 JPEG (which is never shorter than 20,000 bytes)

EXTRACTED BY CARLINI ET AL., 2023

109 images out of 175 million candidates

they selected the 350,000 most duplicated examples and generated 500 candidates for each; the extraction threshold is a hundred duplicates in the corpus

Payment address: RIGHTS.

The model weighs 4.27 GB, trained on 2.32 billion images — 1.84 bytes per picture.

And yet duplicates get memorized: 109 extracted out of 175 million candidates, at a threshold of a hundred duplicates in the corpus. The first half without the second is advertising; the second without the first is an accusation.

the Stable Diffusion 1.5 model card · Carlini et al., «Extracting Training Data from Diffusion Models», 2023 · the LAION corpus downloader tool's report

Legal Case: Getty v. Stability — What the Court Decided, What It Didn't, and What It's Valued At

WHAT WAS DECIDED

the model's weights are not an infringing copy

under UK copyright law; secondary infringement wasn't proven

THE PLAINTIFF'S NARROW WIN

on trademark

the generated images contained artifacts of the plaintiff's watermarks

THE CORPUS: A ×50 SPREAD FROM THE RATE

12.3 million / 5.85 billion pairs · wholesale $2.50 → $30.75 million · retail $130 → $1.6 billion

the real agreed rate is $0,002–0,005/generation (music); the >$9 billion ceiling is theoretical, not awarded

Payment address: RIGHTS.

What the court didn't decide: the lawfulness of training · where it physically took place · the outcome of the appeal.

Inside the corpus — 12.3 million of the plaintiff's images out of ≈5.85 billion pairs. Wholesale $2.50 → ≈$30.75 million; retail $130 → ≈$1.6 billion. A 50-fold spread comes from one choice of rate. The actually agreed rate is three to four orders of magnitude lower.

Five statuses for a legal news item: suit filed · ruling handed down · settlement with no admission of guilt · law in force · law passed, not yet in effect.

the High Court of England and Wales, Mrs Justice Joanna Smith, ruling November 4, 2025 · permission to appeal granted January 2026; no appellate ruling as of 27.10.2026 · stock agency price lists · label licensing agreement terms, 29.10.2025 and 25.11.2025 · denial of rehearing, US, March 2, 2026 · the EU AI Act, applicable from August 2, 2026 · Federal Law No. 243-FZ, in force since 01.09.2026

Map of the Evening: the "Image" Column Filled In

The unit got cheaper by three to four orders of magnitude from any of the three classical baselines. Five boundaries are named by mechanism, not by symptom.

Selection, rights, and reproducibility moved to the right — and separately, the risk of attributing someone else's win to the model.

Тип задачи	Image	*Sound*	*Video*	*3D*	*Recognition*
Generate a first approximation	0,034–0.05 $				
Bring to production quality	adapters and manual touch-up				
Restore what already exists	upscaling the archive				
Hold exact measurements	no: dimensions don't exist				
Fit the real-time budget	not applicable: there's no frame budget				
Answer "who is accountable"	no: the source isn't recorded				

PAYMENT ADDRESS Image — selection, rights

Section 2. Speech Synthesis, Performer Voice, and Music



Sound is the only modality where the section's main claim is checked by the listener's own ear in ten seconds, with no table at all. That is why here the demonstration comes first, the description after.

Three breakdowns, in order: a voice for someone whose disease was taking it away · dubbing and localization · Val Kilmer's voice, given back.

Six mentions, each with a number. The live demonstration plays on the room's speakers.

One voice, two recordings. Which one is real?

THERE IS NO ANSWER ON THE FRAME

the room answers

a raised hand for A or for B

Recording A

Recording B

one and the same person

ten seconds of speech

ten seconds of speech

the answer — a raised hand

Both tracks are the same length and play back to back, with no caption and no pause between them

Two recordings of the same voice are about to play, back to back. One was recorded by a human, the other was built by a model.

What this craft is, and how it was done before the model

FIVE CRAFTS AND THE STAGE'S DECISION

synthesis · cloning · dubbing · music · separation

recognizability and term/territory — what an image does not have

CLASSICAL MODES 1–2 — THE VOICE TALENT AND BROADCAST RIGHTS

$50–450 per minute + $650–6,500 beyond the session

the recording and the right to air it are different purchases

CLASSICAL MODE 3 — LICENSED STOCK MUSIC

$20–100 per track

extended $100–500+; in Russia 10,000–50,000 ₽; a reference point for comparison, not a commissioned score

TWO HONEST GAPS

voice banking · subtitles — no number

no publication names the hours of protocol or the rate per minute

Five crafts, one word. The decision — whose voice and what music the material speaks in.

The classical mode counted this in the price of a finished minute — the same unit as the model.

GVAA Rate Guide, 2026 edition; the SAG-AFTRA agreement on digital replicas

Breakdown: a voice for someone whose disease was taking it away

THE PRICE OF BOTH ERRORS

not in time — the voice is lost forever; recording too early — hours and strength spent in the worst period of a life

after the moment of recording there is no going back; the second error has no money scale — that is a property of the case itself

BOTH CLASSICAL MODES

no source has a number for either

banking: no hours of protocol, no cost of care. A voice out of the box is cheap, but gives a different outcome — someone else's voice

THE MODEL

1–2 minutes of recording for a copy, 1–3 hours for fine-tuning

license $1,200/year, free in the US with this diagnosis. What moved is not the budget but the moment of decision



What moved

What moved is the moment of decision: who holds rights to a voice model after the person is settled by contract, not by law

ElevenLabs Impact Program; Bridging Voice; Scott-Morgan Foundation

Breakdown: dubbing and localization

THE COST OF A RECORDING ERROR

$50–450 + $650–6,500 on top

re-recording after delivery means re-assembling the voice talent

MODE 1 — LIVE VOICE TALENT

$50–450/minute

by experience: $50–150 beginner, from $350 senior

MODE 2 — SUBTITLES

no rate per minute

the mode is named, has no number

MODEL

$0.03–0.072/minute

per 1,000 characters: $0.04 · 0.08 · 0.03 — fast, quality, third provider

THE GAP

700–15,000 times

from the classical mode, not from zero

What the model closed

- Carrying the language with the timbre — a button in the browser
- 70+ languages; intonation — set by tags in the text
- Over a quarter of views are not in the primary language (channel count undisclosed)

What is left to the human

- Timing adaptation: fitting the translation into the line's duration (isochrony)
- Emotion not from text — set by markup
- Lip sync — a separate model, not dubbing
- The cost of 'did not localize': no number anywhere

GVAA Rate Guide 2026; ElevenLabs and Gemini price lists, read 10/27/2026

Breakdown: Val Kilmer's voice, given back

THE PRICE OF BOTH ERRORS

the outcome is binary

either the role sounds in the actor's voice, or it does not sound at all; this is not measured in money

THE CLASSICAL MODE

it does not exist, and the reason is medical

the voice was lost after a tracheotomy for throat cancer, 2014 — a different outcome, not a different price

THE MODEL

more than forty voice models, one chosen

an archive of the actor's recordings, roughly a tenth of the usual data



What moved

The precedent moved: a voice model outlives the person, and from then on a contract controls it, not a right

Sonantic (acquired by Spotify in 2022); "Top Gun: Maverick," 2022

Six observations, each with a number

TRAVIS'S VOCAL

a debut at number 45

11 years with no recording; 20 years off country radio

DARTH VADER'S VOICE

the recordings are forty-five years old

rights transferred during the actor's lifetime — the point of reference for precedent ever since

THE ACCENT IN THE BRUTALIST

only the Hungarian lines were touched

the English untouched; the same service — 3 more nominees that season

"NOW AND THEN"

0 invented notes

45 years from the demo to release; mixing failed in 1995

SIGN-LANGUAGE TRANSLATION ON THE DEPARTURE BOARDS

2,500 stations

the entire UK rail network; the avatars are Deaf people, paid

AN INDUSTRY NORM GOT AHEAD OF THE LAW

human contribution must be 'more than de minimis'

the definition exists in a 2023 award rule; in the laws, as of 2026 — it does not

CNN, 05/06/2024; Variety, 2022; CBC, 2026; The Conversation, 2023; Signapse; Forbes, 06/16/2023

The neural audio codec and residual quantization

input

Convolutional encoder

24,000 samples/s → a vector of 320

75 frames/s = 13.3 ms of latency

the work · the bottleneck

Residual quantization

8 codebooks × 1,024, corrects the error

2^80 flat; 8×1,024 — a table

output

Decoder

tokens → waveform, trained end to end

a minute = 36,000 tokens, 147× less than 5,292,000 on a CD

depth is almost free, codebook width is expensive: 4.01 / 3.98 / 3.92 at 8 / 16 / 80 codebooks

Three units of data, side by side

- A language-model token: a number, vocabulary ~200k
- A diffusion patch: 1024×1024 → 4096, no vocabulary
- An audio token: a number from a codebook, several codebooks

What it costs in volume

- A minute from a CD: 5,292,000 samples
- A codec, 8 codebooks: 36,000 tokens, 147× less
- A music model, 4 codebooks: 12,000 tokens, 441× less
- A minute of speech as text: ≈200 tokens, 180× cheaper

SoundStream; EnCodec; MusicGen

The path from request to result: from text to sound

REPEATS, AND HOW THE REQUEST PARTICIPATES

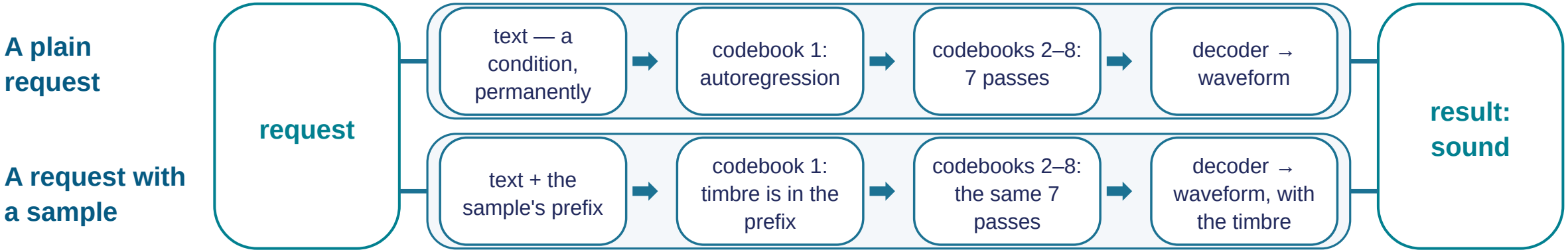
7 passes over the whole frame

from the depth of the codec's cascade (8 codebooks), not from the generator's size; the text is a condition on every pass, not just at the start

WHAT CHANGES, THE BOTTLENECK

codebook 1 — solved once; codebooks 2–8 — refine it

the bottleneck is the codec's own frame rate and cascade depth: a generator is never better than its own decoder



Text holds as a condition across all passes. Codebook 1 is solved once — codebooks 2–8 refine it over seven passes in a row.

VALL-E (Wang et al., 2023) · SoundStream · EnCodec

The codec's frame rate — the main lever on length

NINETY MINUTES OF COHERENT DIALOGUE COMES FROM HERE

7.5 hertz

not from the generator's size. DiffRhythm 2 — 5 Hz, the lowest; Qwen3-TTS — 12 Hz right in the name; Stable Audio 2 — 21.5 against 40–150 in earlier work

CONVERTED INTO VOLUME

a minute of speech — 9,000 to 36,000 tokens

the same minute written as text — about 200 tokens



codec frames per second, hertz

The codec's frame rate is an architectural decision that determines output length; the counterpart of the compression ratio in video and patch size in images.

The main lever is the tokenizer; model size is secondary here.

VibeVoice; Qwen3-TTS; DiffRhythm 2; Stable Audio 2

Autoregression versus diffusion: a measured frame

	a minute of sound	4 min 45 s — 12 min 53 s	grows linearly: every token, its own pass	quality 204.03 and 218.02
Autoregression	two minutes — 6 min 38 s			
Diffusion	two minutes — 13 s	4 min 45 s — 13 s	does not grow: a fixed number of solver steps	quality 79.09 and 81.96

Autoregression: its own task, one text does not have

- Quantization: several streams, a transformer — just one
- Flattened unrolling: 6,000 steps, quality 0.86
- The delay pattern: s, s−1, s−2 — 1,500 steps, 0.96
- A naive all-at-once approach: 1,500 steps, worse to the ear — 2.58

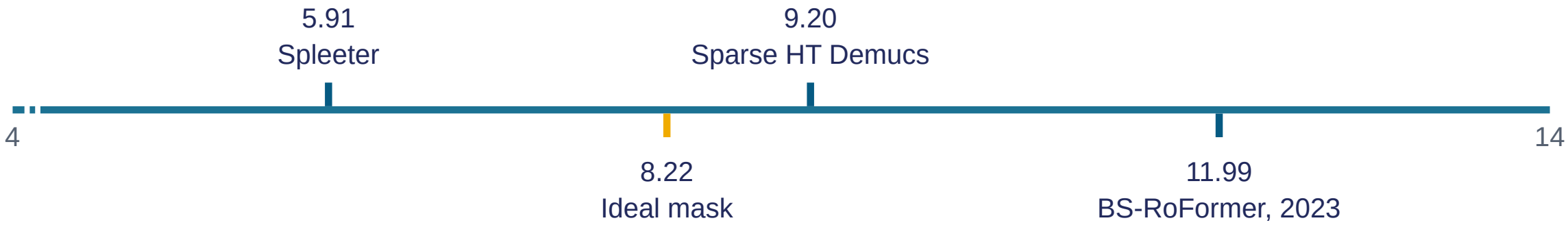
Diffusion: its own task, one text does not have

- A fixed-shape tensor — one single duration
- Time as conditioning: the source's start and length
- A 95 s window; the tail past the needed length is silence
- The tail is trimmed; the 'set before launch' limit remains

The gap is structural: it follows from the design of both families, and the quality of the engineering work has nothing to do with it.

a comparative benchmark of Stable Audio 2 and MusicGen-large-stereo

Voice cloning and source separation are different tools



dB, signal-to-distortion on the vocal · the ideal mask — the masking ceiling; Hybrid Demucs was the first to clear it, at 8.34 dB

The oracle is computed knowing the true tracks

Source separation

- The mix is a sum of tracks, the task is to return the addends
- A ground truth exists: the original isolated tracks
- The model minimizes error against the ground truth
- The oracle is the upper bound when the answer is known

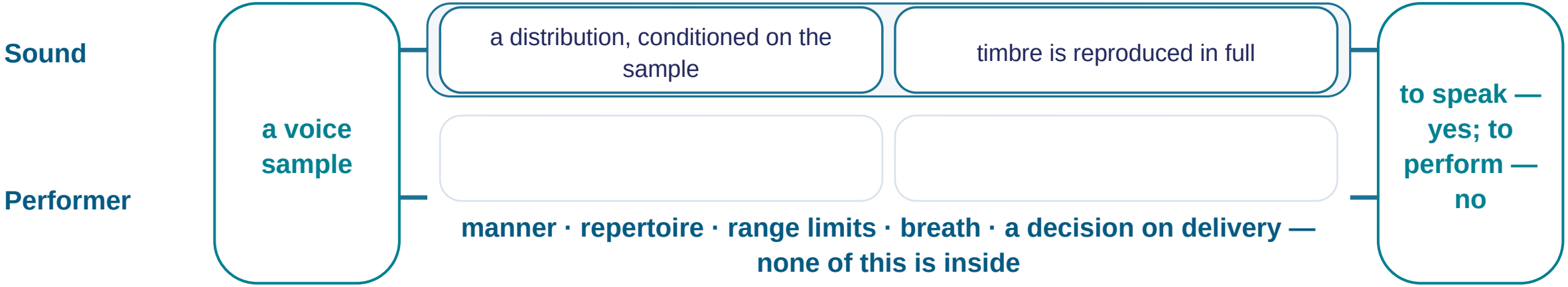
Generation

- No ground truth: there is no 'correct' song
- There is nothing to compare the result against — no metric
- Generation has no ideal oracle at all
- What was recorded is retrieved; inventing is the wrong tool

The boundary between restoration and invention runs along whether an oracle exists.

Spleeter; Demucs; BS-RoFormer; WingNut Films, 2021

The model has no concept of a performer



The root is in the data: the training corpus was assembled from audiobooks

Four consequences, named by the providers

- A clone from calm narration does not carry emotion
- Developers: 'lacks fine control of the details'
- A vocal — a melody with no words, a form with no content
- Cloning of singing is not supported

What it means for tasks where recognizability is what is paid for

- Timbre is recognized; recognizability of a performance — no
- Emotion is manual markup; control exists within its limits always be close

A voice model reproduces a sound; that is where its knowledge of the person ends. The root is in the training data, and the providers name it themselves.

ElevenLabs documentation; F5-TTS; Stable Audio 2

Output length is set before launch: you cannot stop early

WHAT YOU CANNOT ASK FOR

"play until the idea runs out"

you have to name the number of seconds

WORKAROUND 1: DURATION CONDITIONING

paid for with an empty window

a 95 s window; silence in the tail is still counted

WORKAROUND 2: CONTINUATION IN FRAGMENTS

paid for with structure

only the last chunk goes into the context; for one model that is 12 seconds

WHY THE CHORUS DOES NOT COME BACK

there is no reference to the past

a window holding the whole composition — the repeat happens on its own

A language model	length is not set	➡ generation runs	➡ it outputs an end token	➡ length is the result
Diffusion (sound)	length is set as a number	➡ a tensor of that shape	➡ there is nothing to stop it with	➡ length is a parameter

Who decides on the length

The right to one's voice: no protection in the Russian Civil Code

THE INDUSTRY'S POSITION

they are asking for a license, not a ban

the Russian Voice-Over Union: "so that this technology becomes profitable for voice-over artists themselves, rather than the sunset of the profession." No litigated cases under either state law could be found

Russia	September 2024: submitted	January 2025: not supported	October 2026: not adopted
US, states	Tennessee: in force since July 1, 2024	California: since January 1, 2025	no litigated cases found
US, federal	June 2026: approved by committee	September 2026: a senator objected	not passed by either chamber

Three different timelines, and not one of them has ended in a case decided in court

Bill No. 718834-8; the ELVIS Act (Tennessee); AB 1836 (California); the Russian Voice-Over Union

The Andronova case: 600 pages of narration in 2019 → voice in someone else's ad in 2022 → a ₽6 million lawsuit

2019: narration

2019

about 600 pages of narration for a bank; no word 'synthesis' in the contract

2022: voice in an ad

2022

the voice turns up in third-party advertising, available for synthesis on third-party platforms

2023: lawsuit

2023

a lawsuit for nearly ₽6 million, filed August 30 after declining a settlement of two months' pay

2024: ruling

2024

the claim is denied: the court treated the voice dispute as a contract dispute

stage of the only Russian case on a synthesized voice

The only Russian case on a synthesized voice to reach a ruling



Case No. 2-3580/2024, Savyolovsky District Court of Moscow, 08/12/2024; Afisha Daily

A correction to the bill: a downloadable track costs $0.40, not $0.016

THE PRICE OF GENERATION

about $0.016 per track

widely cited; counts a button press

THE PRICE OF A DOWNLOADABLE TRACK

$0.40

eight dollars a month divided by twenty downloads — 8 ÷ 20

THE GAP BETWEEN THE TWO UNITS

a factor of 25

the same service, different denominators

THE GAP AGAINST A $20 –100 STOCK LICENSE

50–250×

not in the thousands, as you get when counting at the price of generation

Тип задачи	Image	Sound	*Video*	*3D*	*Recognition*
Generate a first approximation	$0.034–0.05	dubbing $0.03–0.072/min			
Bring to production quality	adapters and manual editing	manual diction fixes			
Restore what already exists	archive upscaling	source separation			
Hold exact measurements	no: dimensions do not exist here	no: length is set before launch			
Fit the real-time budget	not applicable: there is no frame budget	not applicable: there is no frame budget			
Answer 'who is accountable'	no: the source is not recorded	no: the source is not recorded			

PAYMENT ADDRESS Image — selection, rights · Sound — rights, discernment

a music service's price list, read 10/27/2026

Section 3. Moving Image

The same mechanism as the still image, and two orders of magnitude more expensive: five seconds of high-definition video — **249,240 tokens** against **4,096** for a 1024×1024 image. That is why industrial scale is visible here with the naked eye.

Three breakdowns in order: Chinese micro-dramas · Kalshi, NBA Finals, June 2025 · Coca-Cola's "Holidays Are Coming."

Six mentions, each with a number.



What Kind of Work This Is, and What Decision It Makes

THE STAGE'S DECISION

what the viewer will see in every second

behind it sits the longest chain of obligations in the chapter: rights, union rates, insurance, approvals

BOTH CHAINS ANSWER ONE QUESTION

which shot makes it into the edit

the selection stage hasn't disappeared — it has grown

Classical	Producing the shot is expensive: the shift, lighting, performers, permits	Selection is cheap: the editor watches the footage and chooses	Shot-to-released ratio: 10:1 in advertising, tens-to-1 in documentary
Generative	Producing a shot costs cents	Selecting is expensive: hundreds of clips, character consistency checks, assembling the scene	Trials per usable result — an order to an order-and-a-half more

Where the expensive point sits in each chain

Eleven Production Stages and Nine Ways to Get the Image

THERE ARE NINE WAYS TO GET THE IMAGE, AND NOT ONE OF THE PREVIOUS EIGHT HAS BEEN RETIRED

shooting · location expedition · chroma key · LED volume · motion capture · stock library · archive · 3D rendering · generation

the virtual-production market kept growing by more than thirty percent a year even after the generative wave

Eight stages before 2022

- Principal photography: camera, lighting, makeup, actors
- Editing splices shots together; compositing builds a frame out of layers
- Rotoscoping: cutting an object out frame by frame
- Color grading brings the look together; effects add what wasn't there
- Virtual production and animation — without shooting at all

Three stages since 2022

- Generating a shot: a phrase in, a clip out
- Animating: an image in, motion out
- Extending a scene: building onward from the last frame

Generation didn't retire shooting any more than the volume retired chroma key: the answer is often still "a camera."

How This Was Done Without the Model

COST OF ONE AD, RUSSIA

from 300,000 to 10,000,000 ₽

300–800K ₽ (3–5 people) · 800K–2.5M (with actors) · from 2.5M (10–20 people); an actor — from 500K/day

AN AD AS COMPLEX AS THE GENERATIVE CASES

a crew of 50-plus, a one-year cycle

compared against the campaign in the third breakdown

WHAT ALL THE CLASSICAL MODES SHARE

an attempt isn't free

a day costs a day, a machine-hour costs an hour, a panel costs a panel; that stopped holding in 2022



The unit 'cost of one ad'

- Music: a license 10,000–50,000 ₽, an original score from 150,000 ₽
- The ad's unit: 300,000 ₽ for 30 sec — 10,000 ₽/sec of airtime

The unit 'cost of one episode'

- 2024 micro-drama: >1 million yuan
- The Irishman, 2019: a $125–159 million budget, 1,750 de-aging shots
- The price per shot is the same; there are just enough shots for two films

industry price surveys of the Russian market; The Irishman, 2019; Avid Media Composer, 1989; Toy Story, 1995; StageCraft, 2019

Breakdown: Chinese Micro-Dramas

THE COST OF BOTH KINDS OF ERROR

a miss is 10–20 times cheaper; a missed hit — unchanged

this gap is what the whole industry's restructuring grew out of

BOTH CLASSICAL MODES

a live release >1M ¥ · theatrical box office ≈100B ¥ (2025)

≈$137K per release; box office is the same audience, 120B+ in 2026

THE MODEL

50,000–100,000 yuan per release

≈$7–14K, 10–20 times cheaper than the classical mode, not than zero



VOLUME, PACE, SPEED OF TRANSITION

430,000 releases in eight months, over 90% generated

the July 2026 pace — one generated micro-drama every 36 seconds; the transition took about a year against the fourteen years non-linear editing needed

Caixin, 06/01/2026

Breakdown: Kalshi, NBA Finals, June 2025

CLASSICAL VERSUS THE MODEL, IN MONEY

≈$40,000 · 10,000 ₽/sec → $2,000 · 4–10 ₽/sec

a 95% reduction against the same slot's budget — the NBA Finals, not the Super Bowl

300–400

generations reviewed by one person

Generation

Generation

seconds and cents per attempt

Selection — the bottleneck

Selection — the bottleneck

character consistency checks; a 96% failure rate

15

usable clips



NPR, 06/23/2025; the work's author — PJ Accetturo

Breakdown: Coca-Cola's 'Holidays Are Coming'

THE CAMPAIGN EMPLOYED ABOUT AS MANY PEOPLE AS THE CLASSICAL VERSION

about 100 people

the cycle collapsed from a year to a month — confirmed by the company's own leadership

70,000

clips in 30 days

Generation

Generation

3 studios, 4 models

Selection — the bottleneck

Selection — the bottleneck

what made the final cut — not disclosed

1

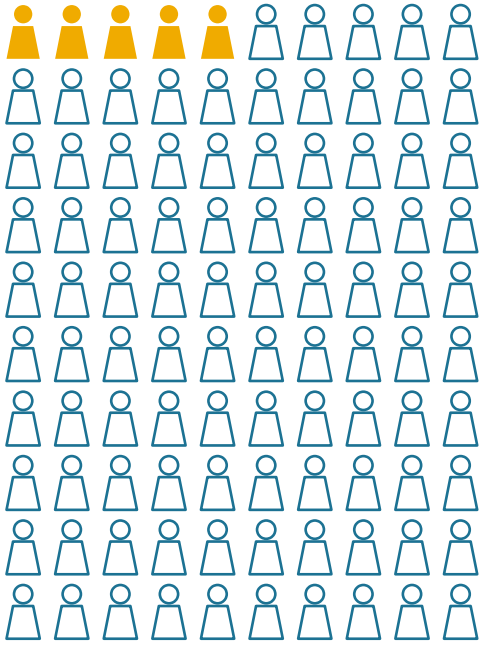
finished ad



Coca-Cola's own behind-the-scenes production video; interviews with the chief marketing officer

The Answer: Just as Many People

100 people on the campaign



5 из 100 pressed the button — generated

95 из 100 selected, assembled, cleared rights — just as many as on a classical campaign of past years

Outcome

Check messages like this with one question: does it name the number of people performing an operation, or the number of people involved in the work?

Two Control Points of Expensive Production

THE CLASSICAL MODE'S BASELINE

half a day per shot

frame-by-frame cutting the object out of the shot — 'a slow, painstaking process'

WITH A BACKGROUND-REMOVAL TOOL

minutes

'I cut it out and drop it on a backing plate'; an intermediate step, done by hand

THE IRISHMAN, 2019

budget $125 → $159 million

1,750 de-aging shots; preparation started 4 years before release

THE CORRECT UNIT

per shot — the same as in an ordinary effects film

it's expensive because there are enough shots here for two films



effects shots in one film · at the Oscar-winning film, an independent five-person studio closed out more than 80% of about 500 shots

Variety, 2023; production materials on The Irishman

Four Thresholds Crossed in Two Years

ON-SET, 2024

budget $45 million

less than The Irishman's increase; criticized for 'plastic' faces

GENERATED SCENE

'10 times faster'

no denominator: 'months,' with no cost given

VISUAL DUBBING

lips matched to English articulation

approved by the same union that struck in 2023

SUPER BOWL LX

30 sec, Feb 9, 2026

20 months from the June 2024 ad

WHAT THIS IS NOT

'half or more of that broadcast's ads'

an industry estimate, not a measurement

Thresholds

2024: de-aging computed on set, film budget $45 million ➡ April 2025: a generated scene makes a final cut ➡ May 2025: a feature film with visual dubbing, approved by the union ➡ February 9, 2026: an ad airs in the Super Bowl's national broadcast

From the first branded ad on a generative model to the national broadcast — twenty months

Variety, 2024; TechCrunch, 07/18/2025; Flawless AI, 05/09/2025; Semafor, 02/04/2026

Spacetime Patches and Spacetime Attention

VIDEO AND IMAGE ARE ONE TASK

'an image is just a video with one frame'

hence training on a mix, and carrying the architecture over

input · bottleneck

Autoencoder

compresses video into a latent

the compression ratio decides duration, resolution, token count

the work

Spacetime patches

the same unit as a text token: a vector in time

arranged into a grid of the needed size

output

Transformer

the patches work as the transformer's tokens

the output size is set by the shape of the patch grid

The parallel is exact: language models have text tokens, video has spacetime patches.

the Sora technical report; CogVideoX

The Cost of the Link Between Frames: 62.1 Billion Pairs Against 2.01 Billion

THE COST OF FULL 3D ATTENTION

62.1B pairs against 2.01B

for five seconds of high-definition video: 249,240 patches, a 31-by-8,040 grid; factorized attention is 30.9 times cheaper

WHAT THAT THIRTYFOLD PRICE BUYS

a direct link between frames

with factorized attention, 'the head of a person in frame i+1 cannot directly attend to the head in frame i' — only indirectly, through background patches

What's the same as in the still image

- The autoencoder is the tokenizer; the model sits on top of it
- Denoising rewrites every token at every step
- No cache exists: language-model pricing economics doesn't carry over

What gets added along with time

- Compression — in both space and time
- Encoding is triple: two spatial axes and time
- 'Full or factorized attention': the price is counted
- The report discloses no details: the figure is a reconstruction

Outcome A model engineering trade-off: consistency bought with compute, and the price paid is known

the Sora technical report; CogVideoX

The Causal 3D Autoencoder

MORE CHANNELS

from three to sixteen, up to 128 at the most aggressively compressing one

compensates for the strength of compression

COMPRESSION RATIOS

4×8×8 at three models · 8×8×8 · 8×32×32

time × height × width; the last one has no patch step at all

input

Input: video frames

the first frame is encoded on its own

the rest are compressed over time

the work · bottleneck

Causal convolution

padding at the start — the future doesn't affect the past

the same trick as an attention mask

output

Latent

frames = the rest ÷ factor + 1

'4/8 plus one': 49, 81, 121, 129, 257

flicker 93.2 → 86.3 when moving to a 3D autoencoder; at 16×16×8, 'convergence becomes extremely difficult'

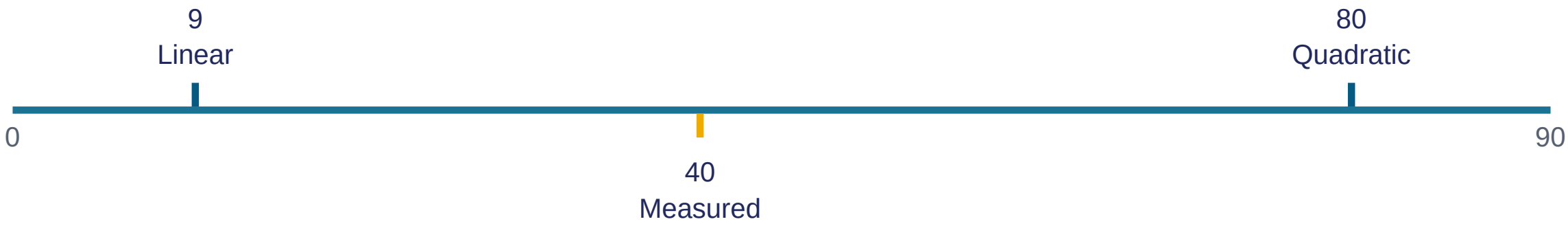
CogVideoX; HunyuanVideo; Wan 2.1; Movie Gen; LTX-Video

Quadratic Growth on Doubling a Frame's Side: the 'Forty-to-One' Arithmetic

THE PRICE LIST REFLECTS THIS

a ten-second clip is three times the price of a five-second one

there's no volume discount here, there's a markup



how many times slower training is when the frame's side triples · 129 frames at 256 points = one time

Outcome Length, resolution, and frame rate are bought non-linearly: threefold by side becomes ninefold by pixels, and attention is quadratic

Open-Sora 2.0; a video-generation provider's price list

The Path from Request to Result: Denoising Spacetime

STEP REPEATS

20–50

one range, with or without a seed frame

WHERE THE REQUEST TAKES PART

at every step

cross-attention holds the vector through the whole cycle

WHAT CHANGES / WHAT STAYS THE SAME

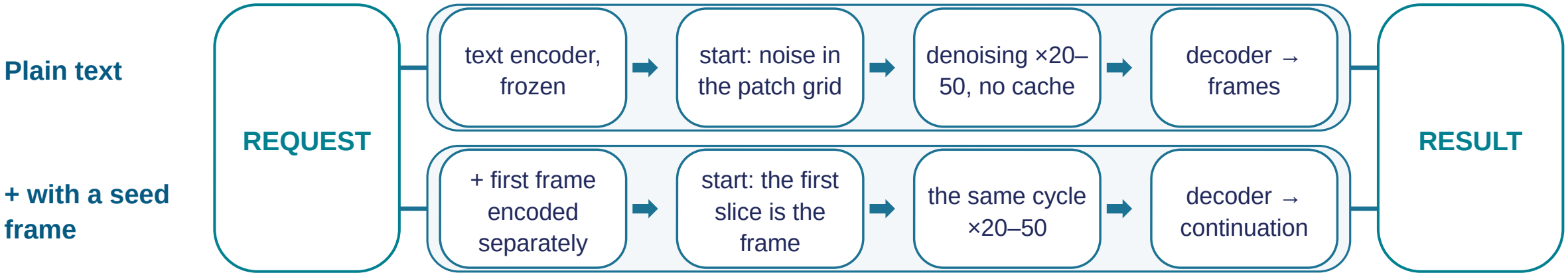
patch values — yes; the grid and the decoder — no

with a seed frame, only the start changes

BOTTLENECK

3D attention with no cache, at every step

factorized attention is 31 times cheaper but loses the link between frames



the Sora technical report; CogVideoX; HunyuanVideo

There Is No Object Between Frames That Stays 'the Same'

1. The shared compression latent	Four neighboring frames are compressed into one latent slice	They don't exist separately	Flicker reduced from 93.2 to 86.3
2. Full 3D attention	The only mechanism with a direct 'object here ↔ object there' link	The cost is counted: thirtyfold	
3. Positional encoding	The coordinate is triple	Channels are split between axes in a fixed proportion	

What holds object consistency together, and where each mechanism runs out

Inside the model there is no entity called "object" that persists between frames: character consistency is the result of four mechanisms at work, and each one is finite.

the Sora technical report; CogVideoX

Where Consistency Breaks Down, by the Developers' Own Admission

WHAT THE DEVELOPERS THEMSELVES ADMIT

'inaccurately models the physics of many interactions'

+ eating food, inconsistencies in long samples, spontaneous appearances of objects

ERROR ACCUMULATION — ACROSS THREE PAPERS

exposure bias

trained on the true context, but run at inference on their own imperfect outputs

PERMANENCE IS A WINDOW SIZE

it's the size of a window, not a property of the world

'memory ~a minute' = 'the cache evicts old frames'

THE ADVICE THAT CARRIES OVER

start with a frame, put the description second

half the uncertainty disappears with one real frame

Exposure bias

During training, the true context is on the input ➡ At inference, the model's own imperfect output is on the input ➡ Error accumulates from frame to frame

object permanence in interactive mode is the size of a window, and it is not a property of the world

the Sora technical report; CogVideoX; papers on exposure bias

Physics Was Learned as Correlation, and It's Been Measured

PLAUSIBILITY AND UNDERSTANDING

coefficient −0.46 at a significance of 0.249

one system's video is the hardest to tell from the real thing

EXACTLY WHERE THE LINE FALLS

handled: paint smearing, pouring liquid

not handled: a ball landing in a box, cutting a tangerine



percent of the upper bound · 396 video recordings, 66 scenarios, three angles each, two takes each

Outcome Plausibility and understanding diverge; sign-off by eye does not work here

Physics-IQ

The Tool Disappears Faster Than Training the Team Pays Off

TWO FLAGSHIPS

within 13 months

the second shut down 10/22/2026

A DEMO DECISION

$800M of expansion halted

a producer's call, on the first examples, 2024

ECONOMICS

$1M/day, <500K users

peak ~1M; not free of charge

THE PROJECT

9 mo. instead of 3 yr — slipped a year

no partner, Q1 2027

2024	February: first examples shown	December: launch for subscribers
2025	Sept 30: 2nd generation + app	Dec 11: $1B / 3yr deal — against 3 unions
2026	Mar 24: the deal falls through	Apr 26/Sept 24: app, access both shut down

The matching answer a successor is named, a migration path is given

a provider's documentation page, read 10/27/2026; reports on Sora's shutdown

The Hidden Cost of Selection

TOYS 'R' US, JUNE 2024

hundreds of iterative shots → a couple dozen

for a 66-second ad — named by a primary source, with the studio's and director's names attached

300

units of source material per one frame that made the cut — 'Air Head,' 2024

The classical mode — the baseline

The classical mode — the baseline

tens-to-one in documentary, about ten in advertising

Selection — the bottleneck

Selection — the bottleneck

the model increased selection by an order to an order-and-a-half against what it imitates

1

frame that made the cut

'Air Head,' April 2024; Toys 'R' Us, June 2024

The Cumulative Map: the 'Video' Column Filled In

The cheapness of an attempt and the expense of choosing grow together: the more expensive the unit of output, the more trials have to be sifted through for one usable one.

Тип задачи	Image	Sound	Video	*3D*	*Recognition*
Generate a first approximation	$0.034–0.05 per image	dubbing $0.03–0.072 per minute	the price of a release is 10–20 times lower		
Bring to production quality	adapters and manual touch-up	manual diction fixes	manual selection and assembly		
Restore what already exists	upscaling archival footage	source separation	frame interpolation		
Hold exact measurements	none: there are no dimensions in the output	none: length is set before launch	none: physics was learned as correlation		
Fit the real-time budget	not applicable: the stage has no frame budget	not applicable: the stage has no frame budget	none: doesn't fit inside a frame budget		
Answer 'who is accountable'	none: the source of the detail isn't recorded	none: the source of the detail isn't recorded	none: the source of the detail isn't recorded		

PAYMENT ADDRESS Image — selection, rights · Sound — rights, discernment · Video — selection

Section 4. 3D Scenes and Real Time

We just counted the ratio of tries to a keeper on video. Here the unit of output is the most expensive in the lineup: the object has to be true from every side at once and in every pose — including poses other people will put it in six months from now, for a different scene.

1 breakdown · 3 mentions · the leading edge. Not one case with both sides costed — and that is information about the modality's maturity.

In order: **Breakdown: "The Wizard of Oz" at Sphere — over 90% of the frame touched**



What this work actually is, and how it was done without a model

REAL-TIME FRAME BUDGET

16.7 milliseconds

at sixty frames per second; geometry, light, shadows, effects, and the interface all have to fit inside it at once

FILM FRAME BUDGET

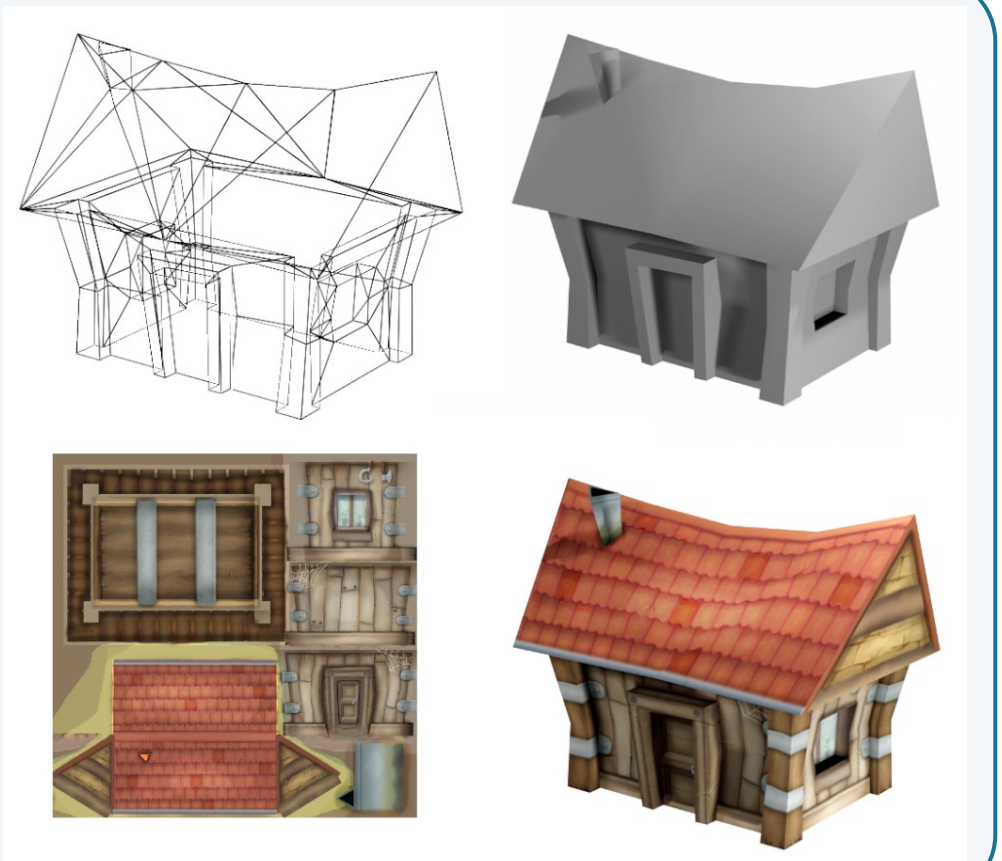
minutes and hours

the same object carries millions of polygons instead of tens of thousands

GAP BETWEEN THE TWO BUDGETS

four to five orders of magnitude

it is what shapes the rest of the pipeline's whole design



The industry holds three different things under the words "3D graphics": the asset, the scene, and the per-frame budget. The stage's decision is to lock in the shape before animation, cloth simulation, and lighting build on top of it.

Mildenhall et al., ECCV 2020, arXiv:2003.08934; QuadGPT, arXiv:2509.21420

The path from request to result: a 3D scene

STEP REPEATS

15,000 (text) · 30,000 (photographs)

not denoising — optimization from scratch

WHERE THE REQUEST TAKES PART

on every one of all the repeats

the judge/comparison holds every step

WHAT CHANGES

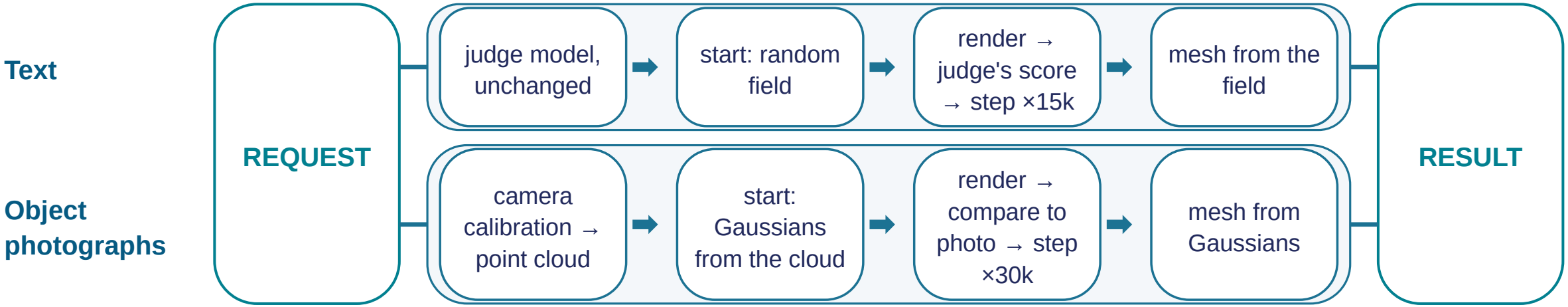
parameters — every step; the representation — no

Gaussians are cloned, split, and deleted

BOTTLENECK

3–4 orders of magnitude more repeats than a ready model

no prior; a ready model takes 20–50 steps



Poole, Jain, Barron, Mildenhall, 2022, Kerbl, Kopanas, Leimkühler, Drettakis, SIGGRAPH 2023 · *arxiv.org/abs/2308.04079*

The classical asset pipeline: what the model touched

HOW DATA ABOUT THE OBJECT IS OBTAINED

photogrammetry · laser scanning · manual modeling · motion capture · ready-made libraries

unchanged, plus a sixth input — text; 7 of 9 stages untouched: retopology, rigging, render…

2 of 9

pipeline stages touched by the model — concept and blocking

Mildenhall et al., ECCV 2020, arXiv:2003.08934; QuadGPT, arXiv:2509.21420

Breakdown: "The Wizard of Oz" at Sphere — over 90% of the frame touched

SOURCE MATERIAL

1939 film stock, 4:3 frame

the breakdown's input

TOUCHED BY THE MODEL

over 90% of the frame

extended on all sides out to the spherical screen

TICKETS SOLD (AUG. 2025 – EARLY 2026)

more than 2.2 million

the 1939 film is available free; people pay for the format

COST OF THE MODES

not published

neither the rights holder, nor the partner, nor any figures outlet has disclosed numbers

Without the model	restoration for a dome, the usual way — no number	➡ 4:3 screening with no extension — rejected, no number either
What it was built with	input 1939, extension by Imagen/Veo/Gemini	➡ upscaling — Topaz; rights — Warner Bros.

Both classical modes are named, but neither is costed

"Not a restoration in the Scorsese sense" — the project's own authors say so. No win is declared at all: both classical modes are named, and neither has a price.

Sphere Entertainment and Warner Bros., project statements, 2025

The LED stage, the asset generator, the described environment

STAGECRAFT LED WALL, OCTOBER 2019

1,326 panels, 270° arc

2.84 mm pixel pitch, 6 m tall, a 23 m floor across; more than half of season one shot inside it

TRIPO GENERATOR: FIRST ASSET OUTPUT

8 seconds on average

no base for comparison: the duration of the same work by a modeler isn't named anywhere, estimates differ by an order of magnitude

GENIE 3: ENVIRONMENT FROM DESCRIPTION

720p, 24 frames/s

environment coherence of several minutes against 10–20 seconds for predecessors of the same class; 11 billion parameters



Generating an image without shooting didn't start in 2024. Nobody was shooting the background of the LED stage back in 2019 either — it was computed in the engine. What changed is how the scene is produced: once it was built by hand, now it's described in words.

Google DeepMind, Genie 3, August 2025 · independent verification WRBench, 23 models, June 2026 (arXiv:2606.20545) · Waymo World Model, February 2026 (demonstration, no independent verification)

NeRF and Gaussian splatting: what got traded for speed

TRAINING ONE SCENE

roughly 70 times faster

48 h for the radiance field against 41 min 33 s for splatting; same scene, same accelerator

RENDERING THE SCENE

roughly 2,200 times faster

0.06 frame/s against 134 — from a research project to an interactive display

MEMORY PER SCENE

roughly 85 times more

8.6 MB against 734 — the very fourth figure that is the price

Radiance field, 2020	scene — one network: point and view → density and color	frame — 150–200 million network queries along a ray	48 hours · 0.06 frames/s · 8.6 MB
Gaussian blobs	scene — elongated, semi-transparent blobs	frame — one projection per blob by a rasterizer	41 minutes · 134 frames/s · 734 MB

The difference is created by the scene's design, not the model's size

Quality didn't change: the difference is under half a decibel, invisible to the eye. The economics changed — and in both directions at once.

Kerbl et al., SIGGRAPH 2023, arXiv:2308.04079; Mildenhall et al., ECCV 2020, arXiv:2003.08934

Generating an object: diffusion straight in 3D versus optimization through rendering

OPTIMIZATION THROUGH RENDERING (DREAMFUSION, 2022)

about 1.5 hours per object

15,000 steps on a four-chip machine, rendering at 64×64 pixels

DIFFUSION STRAIGHT IN 3D (TRELLIS, 2024)

about 10 seconds per object

one forward pass after training on 500,000 objects

GAP

roughly 500 times

both numbers are machine numbers: this is 2022 against 2024 inside one engineering line, there's no human in the pair

Fitting through rendering	model doesn't change — a shape judge	15,000 rendering steps from random angles	output — an object in 1.5 hours
Diffusion in 3D	trained once: 500K objects	one representation → three formats	output — an object in 10 seconds

The same output obtained by two different designs

The gap didn't come from hardware. The first method starts every object from scratch and spends a full cycle of fitting on it; the second trained once on half a million objects — and the cost moved from output into training.

Poole et al., arXiv:2209.14988; Microsoft Research TRELLIS, arXiv:2412.01506

Two scales of one technique: thirty seconds per frame and one millisecond per frame

RADIANCE FIELD, 2020

about 30 seconds per frame

150–200 million queries/frame; one to two days of training per scene

REAL-TIME GRAPHICS, 2025

about 1 millisecond per frame

5 networks/frame, 60 frames/s, a consumer card

BUDGET DIFFERENCE

roughly 30,000 times

the same science: both use a transformer and gradient descent

SHARE OF SCREEN FROM THE NETWORK

15 out of every 16 pixels

at 4x frame generation together with 4x upscaling

What it gives	restores resolution from frame history	synthesizes frames between the rendered ones	3.25 ms → ~1 ms per 4K frame
Where it breaks	doesn't reduce input latency	UI: no motion vectors	glints, particles: motion vectors and depth are wrong

What the network fills in, and where filling in doesn't work

"Can the model do it" is almost always "yes" — and that settles nothing. What settles it is cost, frequency per second, and whose hardware.

Mildenhall et al., ECCV 2020; RIFE, ECCV 2022, arXiv:2011.06294; Waymo World Model, February 2026 (demonstration, no independent verification)

The generated mesh goes in for rework: retopology, unwrapping, rigging

WHAT THE GENERATOR DELIVERS

a first shape in a few seconds

fit for viewing and for placement in the background

WHAT'S INEVITABLY WRONG WITH IT

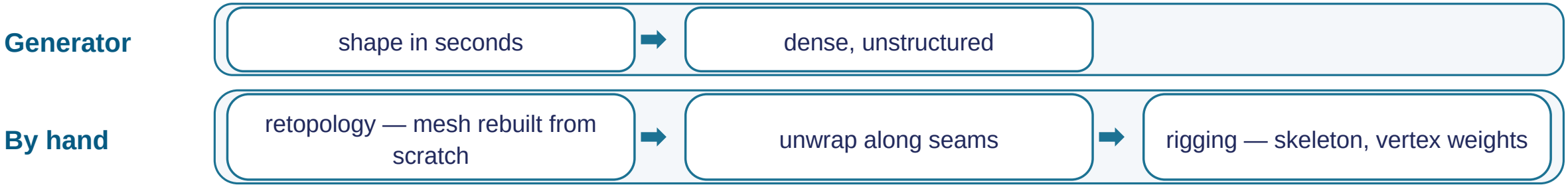
uniform density, no edge flow, no unwrap

follows from the surface-extraction method, not from an unfinished version

HOW MANY STAGES STAY WITH PEOPLE

three of nine — and the most expensive

retopology, unwrapping, rigging; an auto-skeleton doesn't cancel this — bones are set onto the same mesh



What happens to the asset after "eight seconds"

"An asset in eight seconds" and "ready to work with" are different things. The three most expensive stages stay with people.

QuadGPT, Tencent Hunyuan and HKUST, arXiv:2509.21420 (2025)

A generated object has no dimensions

WHAT COORDINATES DOES THE SHAPE LIVE IN

dimensionless

proportions exist, size doesn't

DIMENSIONS FIT FOR A DRAWING

not one

the model is trained on shapes; there were no measurements in training, and there's nowhere for a dimension to come from

OF THE FIVE CLASSICAL INPUTS, THE MODEL TOUCHED

one, and added a sixth

a laser scanner measures size the same way it always has

Shape generator	input — a text description	output — a surface in seconds, dimensionless coordinates	the human assigns scale by eye
Design system	input — parameters and relations between them	dimensions, tolerances, material — part of the model itself	output — a drawing used to manufacture and accept the part

Two tools at the same spot in the pipeline: one has no quantities, for the other quantities are the whole point

Shape comes before calculation — the selection criterion stays with the human. A "load-bearing bracket" looks like a bracket, but carries no load, no material, and no tolerance.

What wasn't found in the open sources

SUBSTANTIVE HITS ON COST

zero

an exhaustive check of the whole collection of materials on prices

WHAT'S MISSING, NAMED

four figures

price of generation per unit of output · modeler's rate · cost of entry into real time · share of the work the asset closes

WHICH MODALITIES HAVE MEASUREMENTS

image since 2023, dubbing since 2024, video since 2025

for 3D, as of end of 2026 — none; it left the lab two years ago

Measurements exist	image — since 2023	dubbing — since 2024	video — since 2025
3D			

not one of the four cost figures — the list is below, in the plates

Three of the evening's modalities have published cost measurements; the fourth has not one of the four figures

This is information, not a gap. Knowing that no measurement exists is more useful than a beautiful number of unknown origin: "a thousand times faster than an artist" would come from dividing a measured figure by a made-up one.

Map of the evening: the "3D" column

Four of five columns. Two rows sit empty across all four modalities in a row: exact measurements and the answer to "who is accountable."

Тип задачи	Image	Sound	Video	3D	*Recognition*
Generate a first approximation *get variations of a look while the decision hasn't been made yet*	$0.034–0.05 per image	dubbing $0.03–0.072 per minute	release cost 10–20 times lower	units of seconds per variant	
Bring to production quality *select, finish, and ship something fit for use*	adapters and manual fixes	manual diction fixes	manual selection and splicing	retopology, unwrapping, and rigging — to people	
Restore what already exists *pull out what's already recorded but reads poorly*	upscaling an archive	source separation	frame interpolation	scene from photographs: 41 min training, 134 frames/s	
Hold exact measurements *dimensions, tolerances, checkable conditions*	no: dimensions don't exist in the output	no: length is set before the run	no: physics is learned as a correlation	no: dimensions don't exist in the output	
Fit the real-time budget *make the frame deadline, not the task deadline*	not applicable: the stage has no frame budget	not applicable: the stage has no frame budget	no: doesn't fit the frame budget	about 1 ms per frame	
Answer 'who is accountable' *provenance, rights, a signature behind the result*	no: the source of the detail isn't recorded	no: the source of the detail isn't recorded	no: the source of the detail isn't recorded	no: the source of the detail isn't recorded	

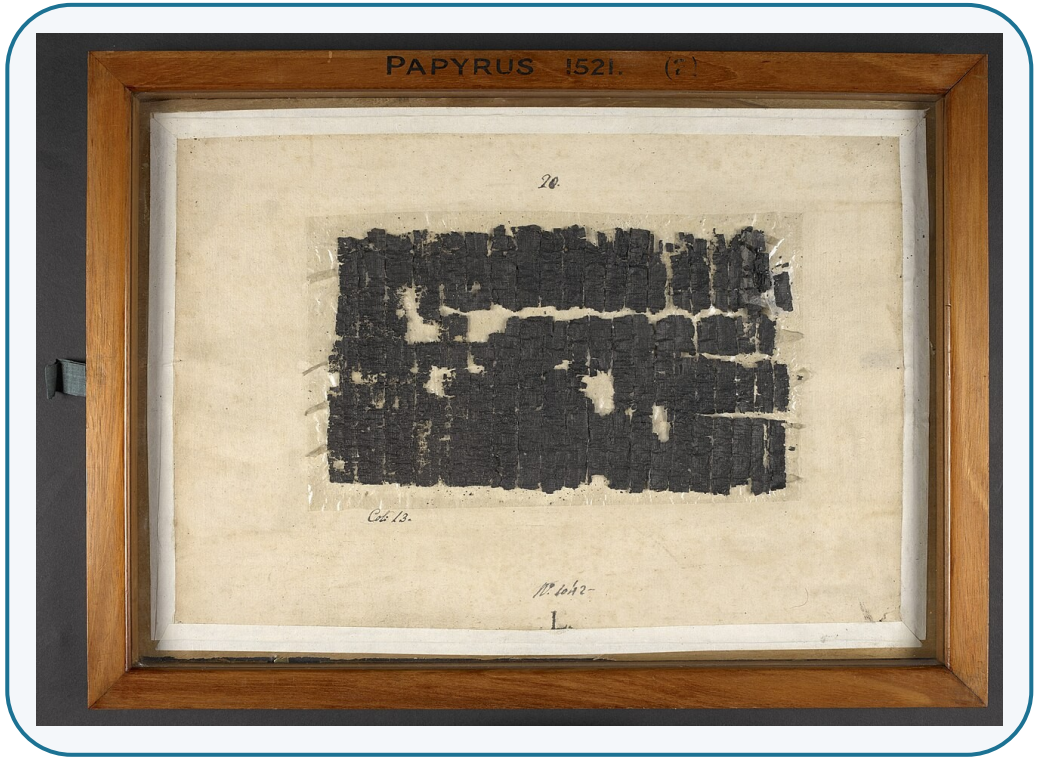
PAYMENT ADDRESS Image — selection, rights · Sound — rights, discernment · Video — selection · 3D — discernment

Section 5. Recognition and Provenance

Four sections in a row, we examined a model that generates. This one reads. The same attention blocks, the same training, often literally the same weights — and an inverse task. From that inversion everything else follows: different economics, different verifiability, different limits.

1 capability breakdown · 2 editorial breakdowns · 2 mentions

In order: **Breakdown: the Herculaneum scrolls — a scroll read in full · Breakdown: CNET — 41 corrections out of 77 articles · Breakdown: Russian media — 18% of articles carrying traces of generation**



What this work is, and how it was done without a model

EDITORIAL FACT-CHECKING — CLASSICAL MODE

40–61% of articles — with an error

independent accuracy audits; this is an ordinary newsroom, with no model at all

OF THESE, PUBLICLY CORRECTED

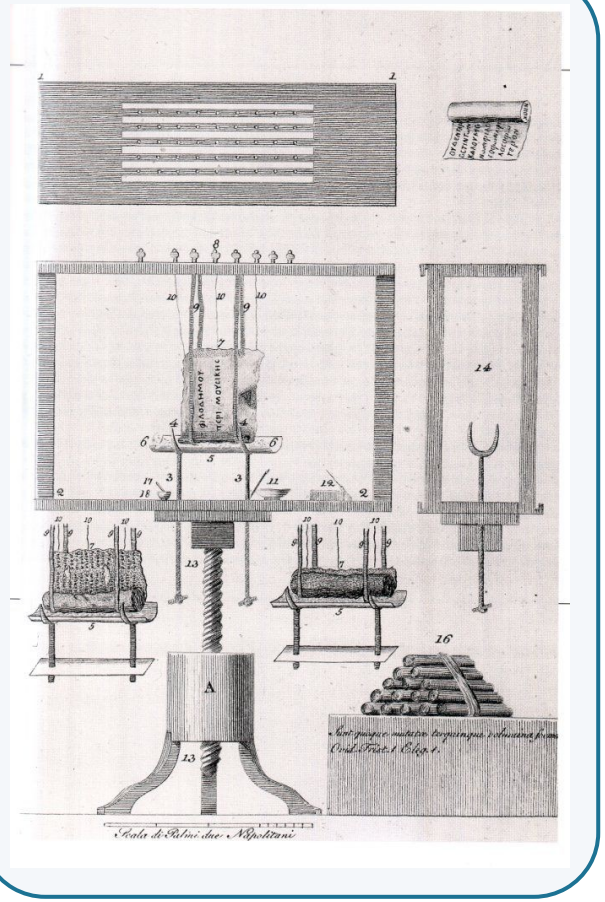
about 2%

that is, every other article has an error, and about one in fifty admits it

AUTHENTICITY EXPERTISE AND MANUAL RESTORATION

weeks and months, no cost figure

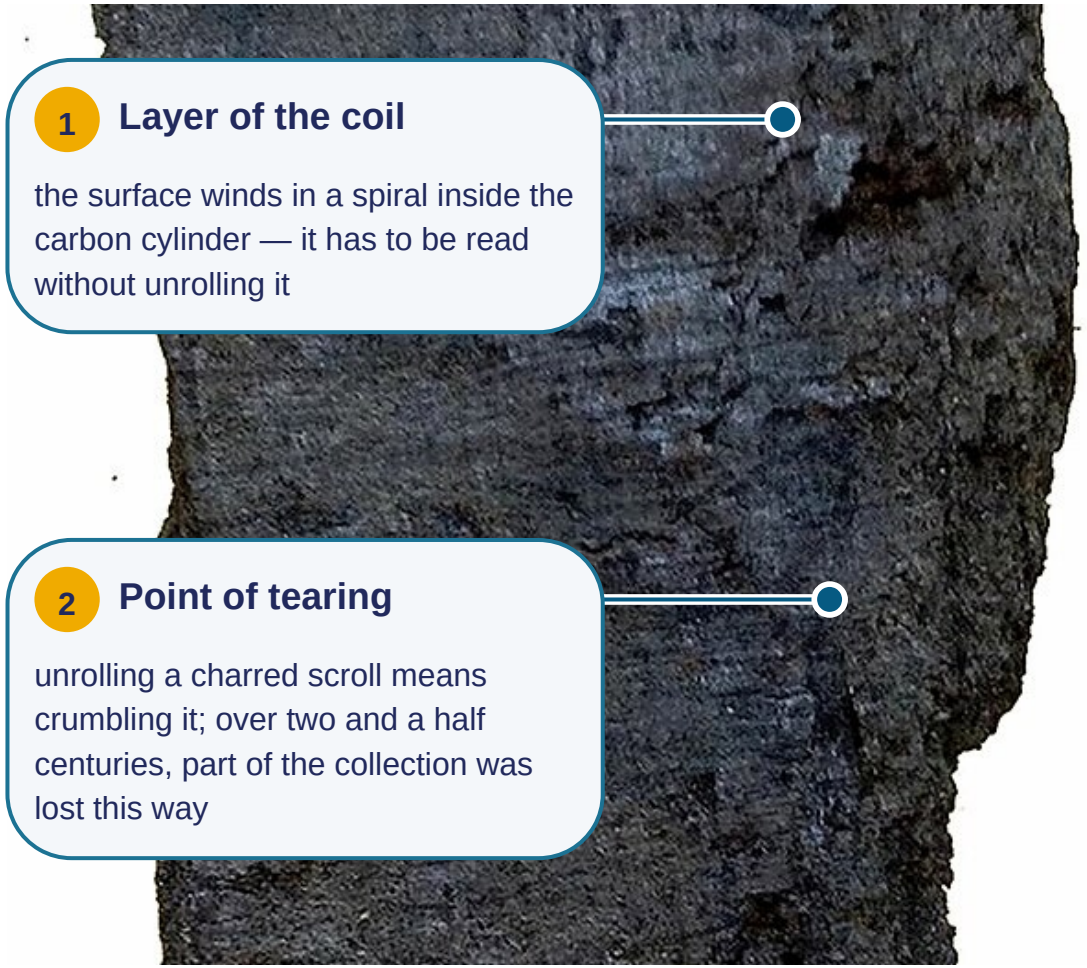
the practice is documented, figures are not published — no multiplier relative to expertise is counted anywhere in the section



Without a model, every other article has an error; it is admitted in every fiftieth. None of the three questions answers "is this true": a genuine photo can lie, a generated one can tell the truth.

independent audits of newspaper accuracy

Breakdown: the Herculaneum scrolls — a scroll read in full



1 Layer of the coil

the surface winds in a spiral inside the carbon cylinder — it has to be read without unrolling it

2 Point of tearing

unrolling a charred scroll means crumbling it; over two and a half centuries, part of the collection was lost this way





3 Ink trace

visible to the eye on an already-opened fragment

Vesuvius Challenge and EduceLab, Prof. Brent Seales, University of Kentucky, in collaboration with the National Library of Naples

How an unrolled scroll is read — and on what dates

How the scroll is read

X-ray tomography of the scroll: layer-by-layer scanning → **digital unrolling: the spiral is flattened into a plane** → **ink search: carbon on the charred base** → a flattened image with the ink revealed

The machine answers one question — where is the ink here; both of its tasks have a checkable answer



milestone in decoding the scroll

How this went by date

Vesuvius Challenge and EduceLab, Prof. Brent Seales, University of Kentucky, in collaboration with the National Library of Naples

The Herculaneum scroll and 'The Wizard of Oz' at Sphere: zero invented against ninety percent



percent of the output frame generated by the model · share of the image that was not in the source data

The technique is a close neighbor, the share invented differs by two orders of magnitude

Herculaneum scroll

- material: charred papyrus, ~1.4 meters
- machine: finds ink on the tomogram
- reads: people, Greek text
- classical mode: does not exist — never read at all

'The Wizard of Oz' at Sphere

- material: 1939 film, 4:3 frame
- machine: fills in the frame on the sides, upscales resolution
- reads: the viewer, as 'that same film'
- classical mode: exists, not counted

Conclusion the boundary — whether the result can be checked independently of whoever obtained it

How a recognizer is built: the same attention blocks, an inverse task

patch → token

Vision model (ViT)

transformer encoder: the same attention blocks

output — a label; orders of magnitude cheaper than generation

image + text → shared space

Pair of encoders (CLIP)

trained on 400 million image–caption pairs

the task — guess the caption for its own image

Generation

the request participates at every one of 20–50 steps; the latent changes, the request's embedding is frozen

Recognition

the request enters once; 12–32 layers in a single pass, the bottleneck is one number

Generation and recognition diverge in three places: cost, verifiability, training.

Dosovitskiy et al., ICLR 2021, arXiv:2010.11929; Radford et al., ICML 2021, arXiv:2103.00020; Nature, October 2024, doi:10.1038/s41586-024-08025-4; Steam's public disclosure pages; platform announcements on dubbing, February 11, 2025

A recognizer for generated content, on an unfamiliar model

AREA UNDER THE CURVE, IN THE SAME ROW OF THE TABLE

94.9

A DIFFERENT measure: 0.5 of 1 is a coin, 1.0 is no misses. Here it's 0.949, that is, 'excellent,' on a decision indistinguishable from a coin. If an acceptance review looked at this figure, it saw an excellent recognizer



percent correct at the working threshold · the same recognizer, different sources of images

Its own family · a foreign family · the same image after processing. Half the scale is a coin

A choice between two honest numbers decides whether a system goes into production.

- ask for the share correct at the working threshold, not the area under the curve
- ask what the recognizer was measured on, and check it against your conditions of use
- the fix is incomplete: the average over the same 12 generators grows from 75.8 to 87.8, while DALL·E 2 stays at 66.9

Corvi et al., ICASSP 2023, arXiv:2211.00680

Bypassing a camera signature: an exploit and a hole in the specification

August 25, 2026 — the StrongBox exploit

- requires root on the device — CVE-2026-43499, an ordinary Linux kernel vulnerability, not camera-specific
- with root, the attacker signs arbitrary data with the real camera's key
- victim — Pixel Camera, C2PA Level 2 certification — a model case, not a weak one
- Google: 'Won't fix (infeasible),' $7,500 — a fix would mean rebuilding the pipeline to run entirely inside the enclave

October 2, 2026 — a hole in the specification

- root and device access not needed — it's the format that breaks, not the phone
- 'excludable byte ranges': the hash excludes the file — the signature and timestamp stand over an empty string
- example: a lottery ticket photographed, the numbers written in AFTER the drawing — the signature still claims the shot predates the drawing
- Neal Krawetz warned in June 2025: neither the specification nor the vendor has been fixed

Two discoveries by one researcher, and they cannot be merged into one: one needs root on the device, the other needs only the file itself.

In both, the cryptography worked as promised: the signature is genuine, the hash checks out. What failed is what the signature doesn't cover.

Conclusion a signature attests WHO signed — not WHAT was in front of the lens

David Buchanan (retr0id), 'C2PA Cameras Do Not Survive Contact With Reality,' personal researcher blog, August 25, 2026 (first discovery) and October 2, 2026 (second); CVE-2026-43499; an independent writeup of the second discovery — workos.com, quoting the primary source

Breakdown: CNET — 41 corrections out of 77 articles

UNIT — ONE FOR BOTH SIDES

share of material requiring correction

a rarity — usually only one side is counted

THE OUTLET'S PIPELINE

41 of 77 articles, i.e. 53%

including an error in a compound-interest calculation

NEWSROOM WITHOUT A MODEL

40–61% of articles with at least one error

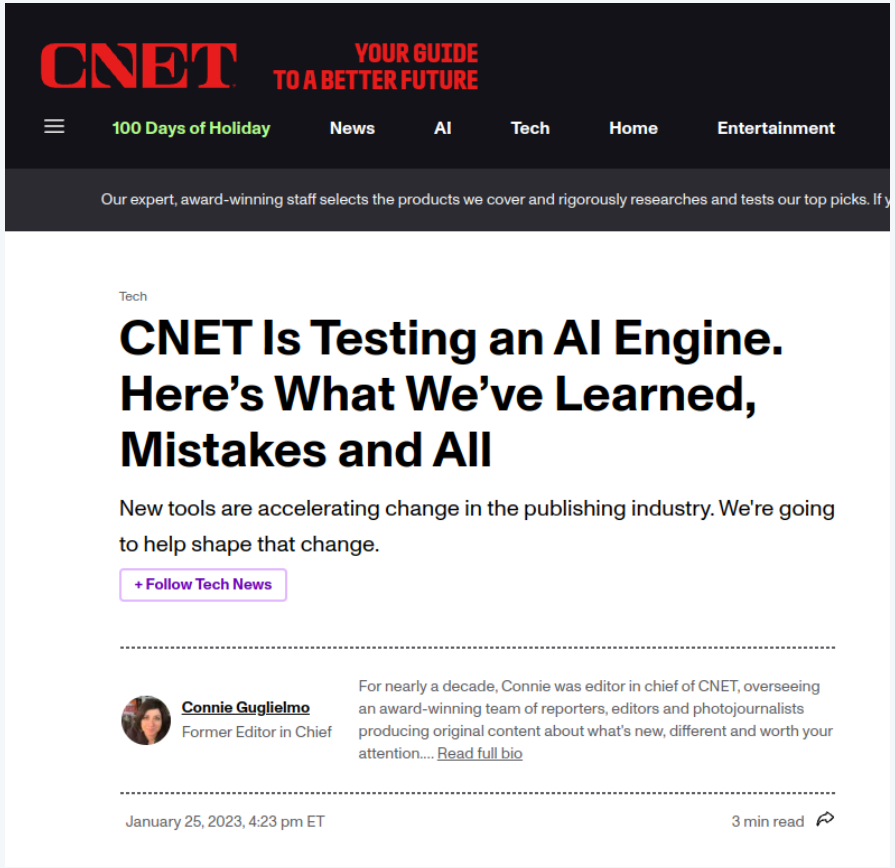
53% from the pipeline falls inside this band

CORRECTED PUBLICLY

about half, versus about 2%

this is where the gap is found — not in errors, but in admission

A new technology's measured failure is compared against the measured behavior of the old one, not against an imagined 'error-free' newsroom. *Both sides, in the same unit.*



CNET, 'CNET Is Testing an AI Engine. Here's What We've Learned, Mistakes and All,' January 25, 2023

Breakdown: Russian media — 18% of articles carrying traces of generation

NEWSROOMS WITH A SHARE OF MACHINE-MADE ARTICLES ABOVE 20%

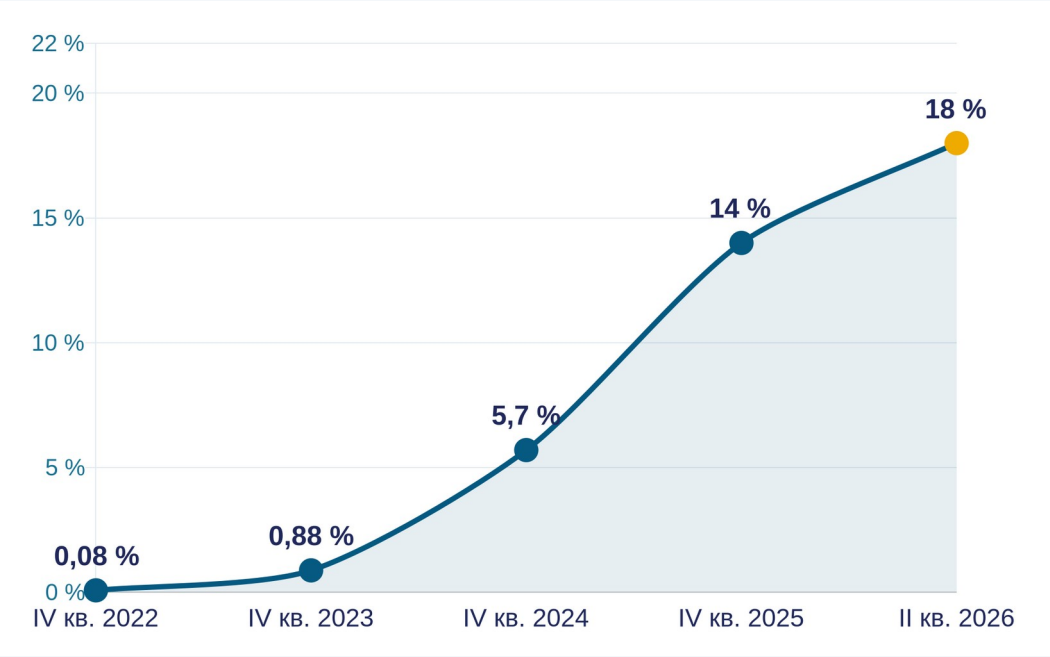
experts per 1,000 words −22%

against their own 2020–2022 level

NEWSROOMS WITH ALMOST NO MACHINE-MADE ARTICLES

+6% over the same period

same corpus, same procedure, same years — this counterfactual is what makes the breakdown the strongest one



The unit — density of live evidence per thousand words. Both halves obtained by one procedure on one corpus: 254,610 articles from 174 outlets, from 2020 through Q2 2026.

Why 18% is a lower bound: recognizers don't generalize to an unfamiliar model. Alongside: 13% growth in machine-like vocabulary among humans, and 27% of the corpus with partial signs.

Kirill Pshinnik, co-founder of Zerocoder, published October 16, 2026

Map of the evening: the 'Recognition' column — the map is complete

Тип задачи	Image	Sound	Video	3D	Recognition
Generate a first approximation *get a range of looks while the decision is still open*	$0.034–0.05 per image	dubbing $0.03–0.072 per minute	release cost 10–20x lower	single-digit seconds per variant	not applicable: generation is not this stage's task
Bring to production quality *select, finish, and release something fit to ship*	adapters and manual editing	manual diction editing	manual selection and editing	retopology, UV unwrapping, rigging stay with people	a human in the loop with a time norm
Restore what already exists *pull out what's already recorded but reads poorly*	archive resolution upscaling	source separation	frame interpolation	a scene from stills: 41 minutes, 134 frames per second	yes: the scroll read in full, the answer is checkable
Hold exact measurements *dimensions, tolerances, checkable conditions*	no: there are no dimensions in the output	no: length is set before the run	no: physics is learned as correlation	no: there are no dimensions in the output	yes: ground-truth labeling gives a number
Fit the real-time budget *make the frame deadline, not the task deadline*	not applicable: the stage has no per-frame budget	not applicable: the stage has no per-frame budget	no: doesn't fit the per-frame budget	about 1 ms per frame	yes: moderation runs continuously
Answer 'who is accountable' *provenance, rights, the signature under the result*	no: the detail's source isn't recorded	no: the detail's source isn't recorded	no: the detail's source isn't recorded	no: the detail's source isn't recorded	partially: a signature outside, and it's about the signer

PAYMENT ADDRESS Image — selection, rights · Sound — rights, discernment · Video — selection · 3D — discernment · Recognition — discernment

Payment address

it accumulates through the same steps — selection · rights · discernment; for both of the section's breakdowns it is one and the same — discernment

Kirill Pshinnik, co-founder of Zerocoder, published October 16, 2026

Section 6. Wrap-up

Five sections went by modality and produced thirteen worked cases. Not one new one gets added here — what gets added is something none of them had on its own: one sentence that can say all of it.

Three addresses · four markets · four refusal criteria. And, separately, where the industry is heading, by the structure of the market: it changes slower than version numbers.



The path from request to result: the same figure, five different starts and bottlenecks

The same figure — five different starts, five different bottlenecks.

Modality	Start	Bottleneck
Image	128×128 latent, 20–50 steps	autoencoder
Sound	codec tokens, 8 books	decoder
Video	space-time patches	autoencoder
3D	points or views, 30,000 steps	scene memory
Recognition	patches, 12–32 layers	one number

consolidated from the device frames accepted for sections 1–5: s16 · s41 · s59/s60 · s68a/s71 · s83

Execution stopped being the scarce resource. What's scarce now — and in what units is it paid for?

WHAT GOT CHEAPER

execution

measured across all five modalities

WHAT DID NOT GET CHEAPER

choice, provenance, confirmation

each one backed by a worked case with measured classic numbers

SELECTION	rejected attempts	300–400 generations → 15 usable cuts: 96% waste	earned by image and video
RIGHTS	money for access to the material	$1.5 billion / 482,460 works ≈ $3,100 per work	earned by sound and image
DISCERNMENT	share of correct calls, the checker's labor	99.9% on its own model, 51.9% on someone else's	earned by recognition

You pay for selection, rights, and discernment: an illustration is only 2.16 times cheaper, against three orders of magnitude of falling execution price — selection stayed expensive.

Catalog share is not money share: four markets that measured independently

WHAT MATCHED ACROSS FOUR PLATFORMS

catalog share is not money share

in different countries, by different methods, across different modalities

Deezer, 2026	over 50% of daily uploads	1–3% of all listens	it was 10% in January 2025
Steam, as of July 2026	30.8% of the year's releases with disclosure	10–27% of estimated sales	it was 3–6% in 2024
US audiobooks, 2025	catalog of 750,000 titles, up 43% in a year	0.03% of sales out of $2.43B	16% had tried a machine voice
Yandex Music, 2026	about 40% of new tracks a month	every tenth chart track	sample of 5.7 million profiles

They measure supply and infer demand — those are different things. Fraud fills the gap: on the music platform, 85% of listens to machine tracks are demonetized.

Deezer Newsroom, July 2026 · Cinevva, July 20, 2026, and an independent count by Totally Human across 53,597 releases · Audio Publishers Association and Edison Research, 2025 year-end figures · an independent measurement across 5.7 million Yandex Music profiles

Where the industry is heading: what limits quality, rights, and market size

IMAGE MODELS STOPPED GROWING

SD v1 (2022) and SD 3 (2024) — the same number of operations

language-model compute grew 4–5x a year over the same period

RIGHTS AND MODALITY — SETTLED, NOT CONTESTED

licensing became a condition of the product; world models are a separate class

Suno and ElevenMusic on licensing since 2026; world-model coherence is minutes, not seconds

FOUR MARKET ESTIMATES FOR 2026

$3.16–19.2 billion — a sixfold spread

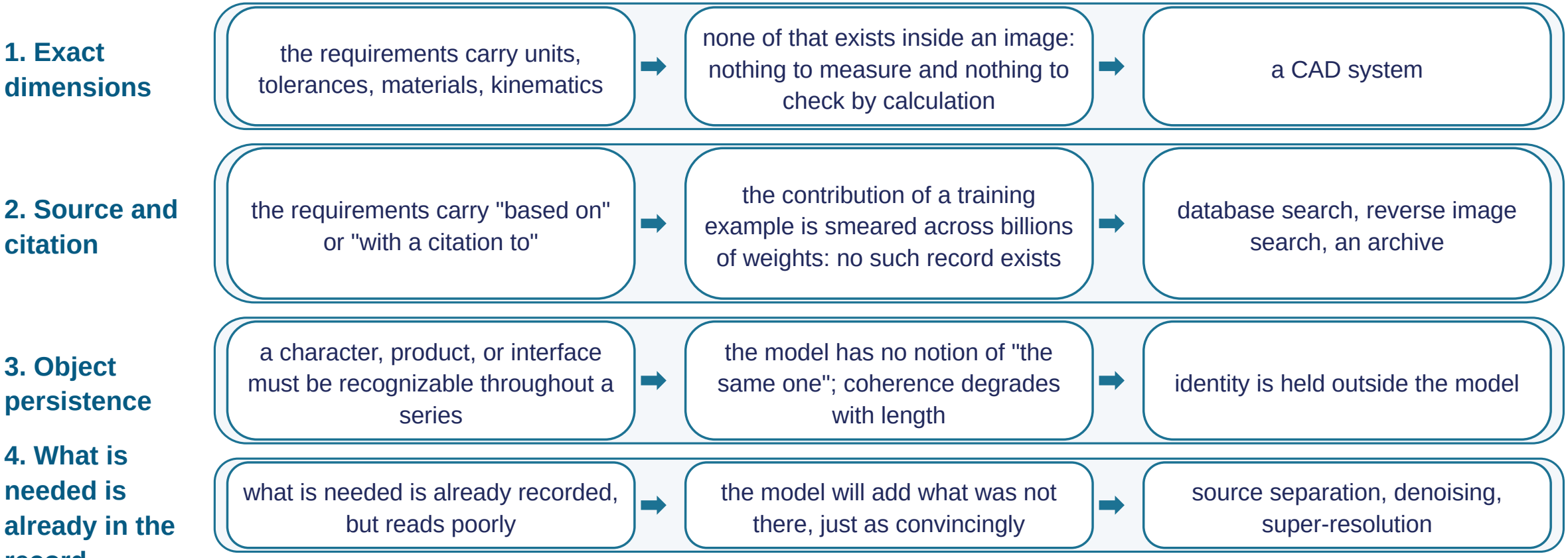
agreeing only on pace: 14–34% a year

The structure of the market changes slower than version numbers. Adoption decisions are better checked against it than against a release number.

Quality is limited by the tokenizer, not model size. An image model has no cache and cannot have one: denoising rewrites all 4,096 tokens at every single step.

SD3 technical report, 2024 · Stable Diffusion v1 model card · Genie 3, Google DeepMind, August 2025 · independent benchmark WRBench, 23 models, June 2026 (arXiv:2606.20545) · Suno v6 release, 09/09/2026, and ElevenMusic, 04/01/2026 · four industry market reviews for 2026

Four refusal criteria: when the model is the wrong tool



The same breakdown for every criterion: sign of the task → why the method cannot take it → what to use instead

Dell'Acqua et al., "Navigating the Jagged Technological Frontier," SSRN 4573321, 09/18/2023 (field study with Boston Consulting Group)

Why criteria get written before work begins

INSIDE THE MODEL'S CIRCLE OF TASKS

12.2% more tasks and 25.1% faster

a large win by the standards of experiments, without the model

THE GAP ON THIS TASK

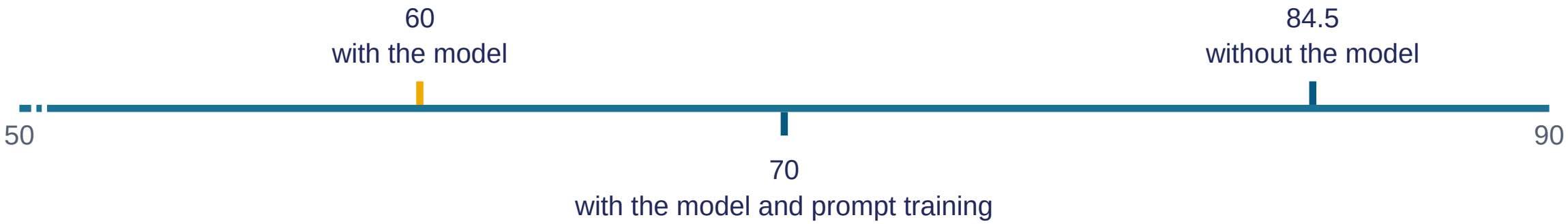
19 percentage points

a difference of SHARES of correct calls, not percent of percent

THE RULE THIS GIVES

criteria get written before work begins

the boundary cannot be seen from inside the task; measured with a stopwatch and a share of correct calls



percent correct on one and the same task · a task outside the circle; 758 consultants, three conditions

The harm comes from the invisibility of the boundary — the quality of the tool has nothing to do with it. The task outside the circle looked no different from the other eighteen.

Dell'Acqua et al., "Navigating the Jagged Technological Frontier," SSRN 4573321, 09/18/2023 (field study with Boston Consulting Group)

Peng vs. METR: the same work, measurements diverging in sign

GAP BETWEEN THE MEASUREMENT AND SELF-ASSESSMENT

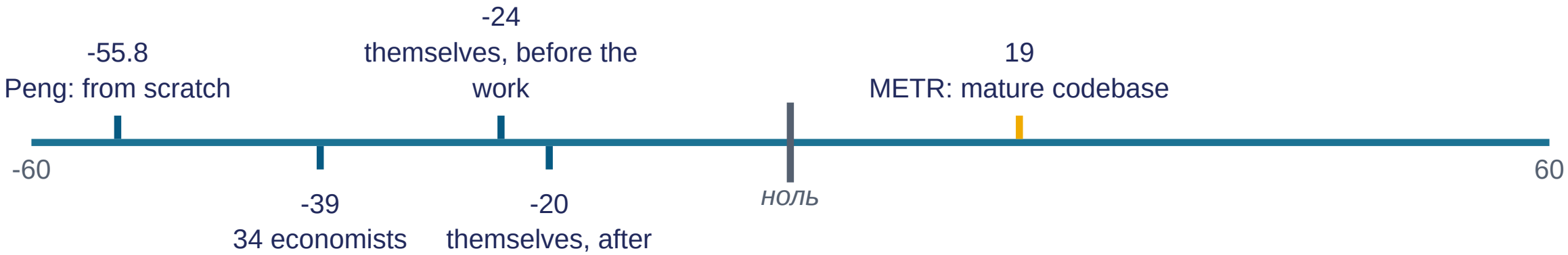
39 percentage points of self-deception

between "19% slower" by the stopwatch and "sure I'm 20% faster" after the work

WHAT TO DO INSTEAD OF A SELF-ASSESSMENT SURVEY

30–50 of your own typical tasks and a stopwatch

for images this is cheaper than for text: evaluating one sample takes seconds



percent of time per task: left — faster, right — slower · the same work without the tool, taken as zero

Four forecasts to the left of zero, the only stopwatch in a mature codebase — to the right

Peng et al., "The Impact of AI on Developer Productivity: Evidence from GitHub Copilot," arXiv:2302.06590, 02/13/2023 · METR, "Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity," 07/10/2025, arXiv:2507.09089

The collapse of diversity: Ju and Aral, 2,234 participants — and the mode in which it doesn't happen

PRODUCTIVITY IN THE FIELD STUDY

50% more ads PER PARTICIPANT

the unit is named: this number does not carry over to a team's total output without recalculation

Lab — 453	time per assignment down 40%, quality up 18%	the draft stage shrank by half, editing doubled	a preregistered randomized study
In the field, 2,234	11,024 ads, checked on real impressions	50% more ads per participant	text wins, image loses

Two studies on the same cheapening of trials: in the lab on standard texts, and in the field on a creative task

The same cheapening of trials, two modes of a human working with a model. The lab study measured time and quality by judges' ratings; the field study measured the behavior of a real audience, with no judges at all. Both found the boundary between parts of the work equally jagged: the text part wins for teams with the model, the image part loses.

Noy S., Zhang W., Science 381(6654):187–192, 07/14/2023, doi:10.1126/science.adh2586 (MIT working paper — 03/02/2023) · Ju H., Aral S., "Collaborating with AI Agents," arXiv:2503.18238, MIT IDE

Two modes of a human working with a model, and the spread collapses in only one of them

The spread collapses	the human delegates the work wholesale and talks to the model as to a conversational partner	the outputs come out noticeably more alike	if the value lies in deviating from the center of expectations, this mode destroys it
It doesn't collapse	the human sets the task, reviews the result, sets the next one	no collapse of diversity is observed in the study at all	both productivity and quality are higher, across every measure at once

The same cheapening of trials, two modes of a human working with a model

The human chooses the mode. That is the only lever on this frame. The first mode is called delegation, the second task-oriented interaction. What distinguishes them is who holds the decision; the tool in both cases is the same one.

The authors' own measure of diversity: seventy thousand variants from one model are less diverse than seventy variants from seventy different people.

Noy S., Zhang W., Science 381(6654):187–192, 07/14/2023, doi:10.1126/science.adh2586 (MIT working paper — 03/02/2023) · Ju H., Aral S., "Collaborating with AI Agents," arXiv:2503.18238, MIT IDE

The count of the throughline: thirteen cases, three unequal parts

ANALYZED IN DETAIL OVER THE EVENING

13 cases

plus one legal case outside this denominator. The addition checks out: 7 + 3 + 3 = 13

Seven, where both sides are measured in the same unit: image price · microdramas · the NBA finals spot · the Christmas campaign · a newsroom with machine-written articles · Russian outlets · dubbing.

Three, where there is no baseline from the classical side, and it is named out loud: how much the same banner would have cost from a staff designer · how many hours of voice banking the clinic's protocol requires · how much a retrofit for a planetarium dome would cost.

Three, where the classical mode does not exist at all: the partition — a shape optimization with no trained model · the actor's voice in the classical mode is either missing or belongs to someone else · for the scrolls there is exactly one alternative, "never to be read."

Zero is a convenient baseline: it always produces an impressive number and is always wrong — somebody did the work before the model. The borderline thirteenth case is named: for dubbing, the first classical mode is counted, the second (subtitles) is not.



What got cheaper wasn't work — what got cheaper was the mistake



year · seven rounds of getting cheaper, each time the intermediary operation disappeared

a first approximation of an idea came to cost fractions of a cent: making seventy thousand variants and picking one became cheaper than making one and hitting the mark

Those who learned to choose, won

- one person: 300–400 generations, two days, the NBA finals
- five specialists in 30 days — a global brand
- microdramas change their production method in a year
- someone who lost their voice gets it back, minute by minute, from old recordings
- a scroll that could not be unrolled — read in full

Those who used to be recognized for free, now pay

- voice, face, signature no longer confirm themselves
- confirmation started to cost money and labor
- 99.9% on a familiar model, 51.9% on an unfamiliar one — a coin flip
- machine-written articles: minus 22% experts per 1,000 words
- the bill for the cheap mistake is not split evenly

A debt owed forward: Lecture 14

WHAT STAYS OFF-FRAME

identity forgery, voice fraud, hiring-process bypass, recognizers inside security systems

a class of attacks — appears when confirmation is no longer free

WHERE THIS GOES

Lecture 14

there the recognizer becomes an element of security: thresholds, the price of an alarm, a human's decision



Execution stopped being the scarce resource — the good news. The work moved onto the step that didn't get cheaper: selection.

Questions